INCH-POUND

MIL-C-49468
14 September 1990

MILITARY SPECIFICATION

CRYSTAL UNITS, QUARTZ, PRECISION,

GENERAL SPECIFICATION FOR

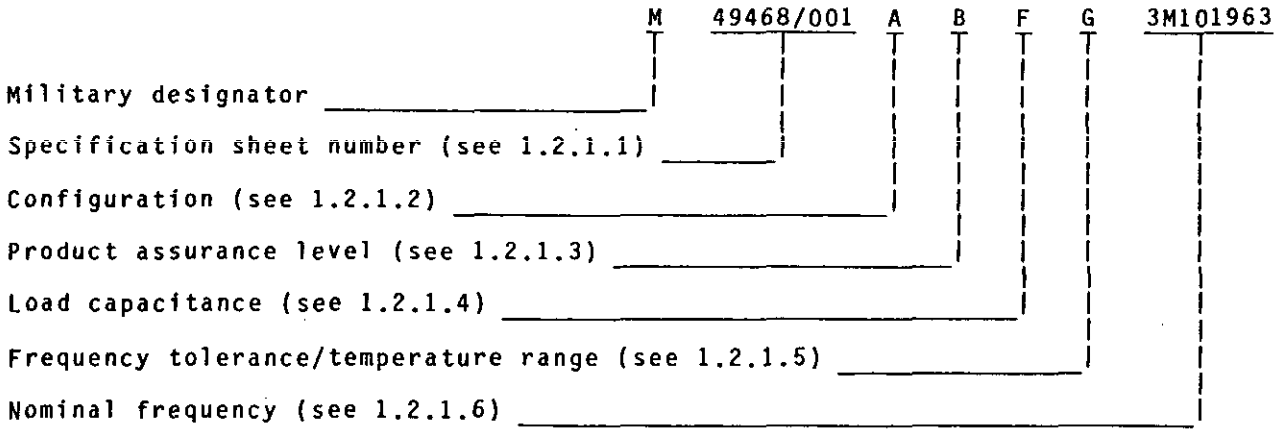
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the general requirements and the quality and reliability assurance requirements for precision quartz crystal units, designed for frequency control in Government electronic equipment.

1.2 Classification

1.2.1 Military Part or Identifying Number (PIN). The PIN shall consist of the letter "M", the general specification number, the detail specification sheet number, a letter designating product assurance level, a letter designating the load capacitance, a letter designating the frequency tolerance/temperature range, and the eight character nominal frequency.



1.2.1.1 Detail specification. The detail specification shall consist of three digits from 001 to 999 as applicable, which identifies a type of crystal unit that has been designed with certain electrical and physical characteristics.

1.2.1.2 Configuration. The configuration shall consist of a single alphabet character which identifies a type of crystal unit that has certain external physical characteristics.

1.2.1.3 Product assurance level. The product assurance level is a single letter, either S or B, designating the device class.

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Research and Development Command, ATTN: SLCET-R-S, Department of the Army, Fort Monmouth, NJ 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5955

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1.2.1.4 Load capacitance. The series load capacitance, which the crystal unit is designed to operate with, shall be designated by one of the following letters.

A	∞ (Series resonance)
B	100 pF
C	50 pF
D	30 pF
E	20 pF
F	15 pF
G	12 pF
H	8 pF

NOTE: The tolerance on the load capacitance should be that value which results in a frequency change not exceeding 10 percent of the frequency tolerance at the reference temperature or 1 percent of the nominal load capacitance, whichever is smaller.

1.2.1.5 Frequency tolerance and temperature range. The frequency tolerance and temperature range shall be designated by a single letter, which is to be selected from a table included in each individual specification sheet.

1.2.1.6 Nominal frequency. The nominal frequency expressed in hertz is identified by a field of eight characters consisting of digits and a letter (H, K, M, or G) representing, simultaneously, the decimal point and the appropriate multiplier as follows.

a. Greater than 1 hertz and less than 1,000 hertz, the letter "H" is used to represent the decimal point and multiplier.

b. Greater than or equal to 1,000 hertz, but less than 1 megahertz, the letter "K" is used to represent the decimal point and multiplier.

c. Greater than or equal to 1 megahertz, but less than 1 gigahertz, the letter "M" is used to represent the decimal point and multiplier.

d. Greater than or equal to 1 gigahertz, the letter "G" is used to represent the decimal point and multiplier.

All digits preceding and following the letter (H, K, M, or G) of the group represent significant figures.

The following are examples of using the eight characters in constructing the frequency.

Compressed notation	Used for frequencies in the noted range
DHDDDDDD	1.000000 to 9.999999 hertz
DDHDDDDD	10.00000 to 99.99999 hertz
DDDHDDDD	100.0000 to 999.9999 hertz
DKDDDDDD	1.000000 to 9.999999 kilohertz
DDKDDDDD	10.00000 to 99.99999 kilohertz
DDDKDDDD	100.0000 to 999.9999 kilohertz
DMDDDDDD	1.000000 to 9.999999 megahertz
DDMDDDDD	10.00000 to 99.99999 megahertz
DDDMDDDD	100.0000 to 999.9999 megahertz
DGDDDDDD	1.000000 to 9.999999 gigahertz

Where D signifies a single digit from 0 to 9.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals
PPP-B-566 - Box, Folding, Paperboard
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-676 - Boxes, Setup

MILITARY

MIL-H-10056 - Holders, Crystal, General Specification for
MIL-P-116 - Preservation, Methods of

(See supplement 1 for list of associated specifications.)

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specification
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of
MIL-STD-810 - Environmental Test Methods
MIL-STD-883 - Test Methods and Procedures for Microelectronics
MIL-STD-1285 - Marking of Electronic Parts
MIL-STD-45662 - Calibration Systems Requirements

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA, 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Defense Logistics Agency

Cataloging Handbook H4 - Commercial and Government Entity (CAGE)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY

NIST Technical Note 1337 - Characteristics of Clocks and Oscillatiors

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

EIA-477 - Cultured Quartz
EIA-512 - Standard Methods for Measurement of the Equivalent Electrical Parameters of Quartz Crystal Units, 1 KHz to 1 GHz

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC-302 - Standard Definitions and Methods of Measurement for Piezoelectric Vibrators Operating Over the Frequency Range up to 30 MHz.
IEC-444 - Basic method for the Measurement of Resonance Frequency and Equivalent Series Resistance of Quartz Crystal Units by Zero Phase Technique in a Pi-network.

(Application for copies should be addressed to the American National Standards Industries, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E595 - Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials From Outgassing in a Vacuum Environment.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references (except for related associated detail specifications, specification sheets, or MS standards) cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Product assurance requirement. Two levels of quality assurance, classes B and S are provided for in this specification. Classs B and S crystal units shall be those which have been subjected to and passed all applicable requirements, tests, and inspections specified herein.

3.2.1 Reliability assurance program. A reliability assurance program for crystal units furnished under this specification as specified in 4.1.2 shall be established and maintained.

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3.3 First article inspection. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.6 and 6.3).

3.4 Qualification. Crystal units furnished under this specification as class B or S shall be products which are qualified for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.7 and 6.4). If there is no QPL for this item, first article inspection shall be invoked (see 3.3).

3.5 Materials. Materials shall be used which will enable the crystal units to meet the requirements of this specification. When required by the contract (see 6.2), the manufacturer shall furnish certification supported by verifiable data that materials used in fabricating the crystal units are in accordance with the specified requirements, and are the same materials used in fabricating the qualification samples.

3.5.1 Outgassing. (when specified see 3.1) Materials utilized shall have a maximum total mass loss of 1.0 percent and a maximum Collected Volatile Condensable Material (CVCM) content of 0.1 percent when tested in accordance with ASTM E595. No cadmium or zinc plating shall be used in the fabrication of the crystal unit.

3.5.2 Holder-base and cover. The crystal holder shall be of the type specified and the materials and presealed physical dimensions specified in MIL-H-10056 (see 3.1).

3.5.3 Quartz. If the type of quartz is specified (see 3.1), it shall conform to the following requirements (see 3.5).

3.5.3.1 Cultured quartz. If cultured quartz is specified, it shall have been cut from a bar which met the requirements of EIA Standard EIA-477. The Q-grade shall be as specified (see 3.1) but not worse than grade 1.8 for thickness-shear type resonators (see 6.5.11). For class S, the grower, autoclave ID, date code, and bar ID shall be provided.

3.5.3.1.1 Swept cultured quartz. If swept cultured quartz is specified, the sweeper, sweep lot number, electrode description (material and application technique, e.g., pressure, sputtered, or evaporated), sweeping atmosphere, sweeping temperature, electric field and current profiles shall be provided as specified (see 3.1).

3.5.3.2 Natural quartz. If natural quartz is specified (see 3.1), documentation must be provided to substantiate the supplier.

3.6 Design and construction. Crystal units shall be of the design and construction specified (see 3.1). The external dimensions shall be as specified (see 3.1) after assembly and marking. Other materials and processes may be used if prior approval is obtained from the qualifying activity.

3.6.1 Supporting structure. Interference, friction, crimped or similar joining of parts unreinforced by solder, welding, adhesive, etc. shall not be used to support or provide electrical contact for the quartz resonator.

3.6.2 Bonding. Electrical connection shall be made to the quartz resonator electrodes by an electrically conductive cement, solder, thermocompression bond, ultrasonic bond, parallel-gap weld, electroplated metal bond or by another method providing intimate metal-to-metal continuity. Interference, friction, crimped or similar joining of parts unreinforced by solder, welding, etc. shall not be used.

3.6.3 Sealing conditions. Crystal units shall be hermetically sealed in vacuum or with a backfill gas as specified (see 3.1). The type of gas, purity, temperature and pressure used at sealing shall be provided for informational purposes.

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3.6.4 Glass parts. Glass seals in the base of crystal units with metal holders shall be free of cracks. Crystal units with glass holders shall contain no cracks in the glass structure (see 4.9.2.1.1). Minute flaking around the feather edge of a meniscus shall not be considered a crack.

3.6.5 Final frequency adjustment. Final frequency adjustment shall not be accomplished by means of abrasion of the electrode or exposure of the crystal resonator to a halogen vapor.

3.7 Solderability or lead attachment.

3.7.1 Solderability. Wire-lead terminals shall be solderable when tested as specified in 4.9.3.1. The criteria for evaluation of solderability shall be as specified in method 208 of MIL-STD-202.

3.7.2 Ceramic package. The terminal pads of the package shall be suitable for the specified lead attachment method (see 3.1) when tested as specified in 4.9.3.2.

3.8 Resistance to solvents. The marking shall not become illegible or discolored when testing as specified in 4.9.4.

3.9 Resistance to soldering heat (when specified see 3.1). When crystals are tested as specified in 4.9.5, there shall be no damage which would adversely affect normal operation of the crystal. The changes in frequency and equivalent resistance shall not exceed the specified limits.

3.10 Frequency and resistance.

3.10.1 Resistance versus temperature (R vs T). When tested as specified in 4.9.6.1, the resistance shall not exceed the specified maximum at any temperature between and including the specified limits (see 3.1).

3.10.2 Frequency versus temperature (static). The requirements for frequency versus temperature depend upon the intended application. The specified parameters shall be within the limits specified when tested as specified in 4.9.5.1 at the drive level, load capacitance, and rate of change of temperature specified (see 3.1).

3.10.3 Coupled modes (frequency - resistance anomalies). The frequency versus temperature characteristic shall be free of the effects of coupled modes over the specified temperature range, as demonstrated by the absence of deviations in the frequency versus temperature characteristic greater than the specified amount from the specified polynomial (see 4.9.6.2.1) when tested as specified in 4.9.6.2. The resistance and, if specified, the slope of the resistance versus temperature characteristic shall not exceed the values specified over the temperature range specified (see 3.1) when tested as specified in 4.9.6.2.

3.11 Capacitance.

3.11.1 Capacitance, shunt. The shunt capacitance shall be as specified (see 3.1), when measured as specified in 4.9.7.1.

3.11.2 Capacitance, motional. The motional capacitance shall be as specified (see 3.1), when tested as specified in 4.9.7.2.

3.12 Quality factor ("Q"). The quality factor shall exceed the value specified (see 3.1), when tested as specified in 4.9.8.

3.13 Unwanted modes. When tested as specified in 4.9.9, the resistance on any unwanted mode within the frequency range shall be greater than the value specified (see 3.1).

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3.14 Insulation resistance. The insulation resistance shall not be less than 500 megohms when tested as specified in 4.9.10.

3.15 Accelerated aging. When tested as specified in 4.9.11, the difference in frequency between the measurement made immediately prior to and immediately after conditioning shall not exceed the value specified. The resistance shall not change by more than the value specified (see 3.1).

3.16 Drive sensitivity (of frequency and resistance). The difference between the maximum and minimum frequency and resistance shall not be larger than the specified amounts (see 3.1), when tested as specified in 4.9.12.

3.17 Condensibles. The dew point of the gas in the sealed holder shall be -55 degrees C or lower as demonstrated by the absence of resistance excursions exceeding the limits specified (see 3.1), when tested as specified in 4.9.13.

3.18 Shock. When crystal units are tested as specified in 4.9.14, changes in frequency and resistance shall not exceed the specified values (see 3.1).

3.19 Vibration, acceleration and acoustical noise.

3.19.1 Frequency and resistance offset (vibration). When tested as specified in 4.9.15.1, changes in frequency and resistance shall not exceed the specified values (see 3.1).

3.19.2 Acceleration sensitivity (vibration). If specified, the vibration induced acceleration sensitivity shall not exceed the value specified for the maximum acceleration specified over the frequency range specified when tested as specified in 4.9.15.2. No mechanical resonances shall be present at any frequency between the limits specified (see 3.1).

3.19.3 Frequency and resistance offset (steady state acceleration). If specified, change in frequency and equivalent resistance shall not exceed the values specified (see 3.1), when tested as specified in 4.9.15.3.

3.19.4 Acceleration sensitivity (steady state). If specified, the frequency change per unit of acceleration shall not exceed the value specified at any acceleration level below the maximum specified (see 3.1), when tested as specified in 4.9.15.4.

3.19.5 Acoustical noise. If specified, the ratio of the power in a 1 Hz bandwidth of a single sideband to the power in the carrier shall not exceed the values specified for the acoustic frequencies specified (see 3.1), when tested as specified in 4.9.15.5.

3.20 Thermal frequency repeatability. If specified, the maximum frequency difference among any of the frequencies at each turnover temperature shall be less than the value specified (see 3.1), when tested as specified in 4.9.16. If there are two turnover temperatures, the maximum difference among the values of f(UTP)-f(LTP) shall be less than the value specified (see 3.1).

3.20.1 Thermal frequency hysteresis. If specified, the thermal frequency hysteresis, as measured as specified in 4.9.16.2, shall not exceed the specified value (see 3.1).

3.21 Particle impact noise detection (PIND). If specified, crystal units shall be subjected to the test specified in 4.9.17.

3.22 Moisture resistance. The frequency and resistance shall not change by more than the values specified (see 3.1) when tested as specified in 4.9.18. There shall be no evidence of excessive corrosion.

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3.23 Salt spray. There shall be no evidence of excessive corrosion when tested as specified in 4.9.19. Corrosion that causes impairment of the electrical or mechanical performance of the unit shall be considered excessive.

3.24 Terminal strength.

3.24.1 Terminal pull. When tested as specified in 4.9.20.1, there shall be no evidence of damage to the terminal or glass seal, or movement of the terminal relative to the glass at the point of the seal.

3.24.2 Wire-lead bend (applicable to crystal units with wire-lead terminals). When tested as specified in 4.9.20.2, there shall be no severing of the terminals, or cracking or chipping of the glass.

3.24.3 Wire-lead twist (applicable to crystal units with wire-lead terminals). When tested as specified in 4.9.20.3, there shall be no severing of the terminals, or cracking or chipping of the glass.

3.25 Bond strength. If specified, the junction between each supporting wire or other supports and the quartz blank shall have the minimum bond strength specified (see 3.1), when tested as specified in 4.9.21.

3.26 Thermal shock. There shall be no evidence of cracking, chipping or breaking in any part of the crystal unit when examined as specified in 4.9.22. If specified, changes in frequency and resistance shall not exceed the specified values (see 3.1) when tested as specified in 4.9.21.

3.27 Thermal time constant. If specified, the thermal time constant shall be defined as the time require for the frequency to reach its "static" value (see 3.9.2) at 63°C ±1°C after a step change in ambient temperature from 0°C to 100°C. When measured as specified in 4.9.23, the thermal time constant shall not exceed the value specified (see 3.1).

3.27.1 Frequency overshoot. If specified, the frequency overshoot, as defined in 4.9.23.1, shall not exceed the value specified, (see 3.1).

3.28 Barometric pressure (reduced). If specified, crystal units shall be subjected to the test specified in 4.9.24.

3.29 Seal. When tested as specified in 4.9.24, the leakage rate of the crystal unit enclosure shall not exceed the value specified (see 3.1) or 1×10^{-8} atm cm^3/s of air, whichever is smaller.

3.30 Short term stability. If specified, the short term stability, as defined in 4.9.26, shall not exceed the values bounded by the straight-line segments drawn on log-log axes through the specified points (see 3.1).

3.31 Aging. When tested as specified in 4.9.27, the total frequency change shall be less than the value specified. If specified, the aging rate per day at day 30 and the maximum frequency change for a one-day period shall be less than the value specified. The resistance shall not change by more than the specified amount (see 3.1).

3.32 Radiation hardness.

3.32.1 Total dose. If specified, the radiation-induced changes in frequency and resistance shall not exceed the value specified (see 3.1), when tested as specified in 4.9.28.1.

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3.32.2 Dose rate. If specified, the maximum radiation-pulse induced changes in frequency and resistance shall not exceed the values specified (see 3.1), when tested as specified in 4.9.28.2. The radiation pulse-induced changes in frequency and resistance after the specified time interval shall not exceed the values specified (see 3.1).

3.32.3 Neutrons. If specified, the neutron-induced changes in frequency and resistance shall not exceed the values specified (see 3.1), when tested as specified in 4.9.28.3.

3.32.4 Accumulated time error. If specified, the time integral of the radiation-induced frequency change for the specified time interval (see 3.1), shall not exceed the value specified when tested as specified in 4.9.28.4.

3.33 Marking. The military part number, including specified frequency, and manufacturer's identification code shall be marked on the crystal at the locations specified (see 3.1). The part number specified in 1.2.1 shall not be marked on any crystal unit for which the contractual requirements deviate in any way from the requirements specified for the crystal unit (see 6.2). Any omitted, relaxed, or added contractual requirement shall be considered a deviation. Crystal units shall also be marked with the year and week of the final test in accordance with MIL-STD-1285. The manufacturer's identification code shall be listed in Cataloging Handbook H4/H8 "Commercial and Government Entity (CAGE)" (see 6.9). The specified frequency shall be marked in compressed form (see 1.2.2). The military part number, frequency and manufacturer's code designation characters shall not be less than 3/64 inch in height unless otherwise specified, arranged as illustrated on the specification sheet (see 3.1), and located symmetrically with respect to the center axis of the crystal holder.

3.34 Workmanship. Crystal units shall be processed in such a manner as to be of uniform quality and free from any defects that would adversely affect life, serviceability, or appearance. The interior of the crystal unit shall contain no flux, particulate matter, residue or other foreign or unapproved materials. There shall be no evidence of fractures in the resonator nor shall there be any cracked or flaked edges. Electrodes shall be clean and untarnished (see 4.9.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Responsibility for conpliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with the applicable requirements of MIL-STD-790.

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4.1.3 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspections shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3.1 Test Sets.

4.1.3.1.1 Reference standard test set. The reference standard test set shall be furnished by the contractor. The reference standard test set shall conform to IEC-444 or EIA-512. The contractor is responsible for certifying to the qualifying agency that the reference standard test set conforms to the appropriate standard (see 4.1.3.1.1.1).

4.1.3.1.1.1 Certification of reference standard test sets. Certification of compliance with IEC-444 or EIA-512 shall consist of the following:

a. Functional diagram of the contractor's system showing interconnections, equipment manufacturer, and model designations.

b. Flow chart and description of all software modules used in the control of equipment and estimation of parameter values.

c. Flow charts and descriptions of instrument calibration and error-correction routines.

d. Traceability documentation for test fixtures and reference impedance devices required for the calibration, verification, and use of the standard test set.

4.1.3.1.2 Contractor's equivalent test set. The contractor's equivalent test set shall be furnished by the contractor, and shall be used for the performance of quality conformance inspection. It is not a requirement that the equivalent test set be identical to the reference standard test set only that the frequency and resistance be correlated prior to performing quality conformance inspection (see 4.1.3.1.2.1 and 4.1.3.1.2.2). The contractor's equivalent test set must be set to operate at the same drive current as the reference standard test set during correlation. In addition, the contractor's equivalent test set shall be designed to permit remote connections to the crystal unit since it is not intended that the test set be subjected to the ambient temperature range specified for the crystal unit.

4.1.3.1.2.1 Frequency correlation. The frequency of a given crystal unit measured in the contractor's equivalent test set (see 4.1.3.1.2) shall be within 0.00001 percent (0.1 ppm) of the frequency (see 3.1) measured in the reference standard test set (see 4.1.3.1.2).

4.1.3.1.2.2 Resistance correlation. The equivalent resistance of a given crystal unit, as measured in the contractor's equivalent test set (see 4.1.3.1.2) shall be within 5 percent or one ohm whichever is greater, of its specified maximum resistance as measured in the reference standard test set (see 4.1.3.1.1).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. Screening (see 4.5).

c. First article inspection (see 4.6).

d. Qualification inspection (see 4.7).

e. Quality conformance inspection (see 4.8).

4.3 Materials inspection. Materials inspection shall consist of certification from the source that the materials used in fabricating the crystal units are in accordance with the applicable requirements prior to such fabrication. In the absence of certification from the source, a certificate of analysis or certified inspection data shall be required as proof of conformance to applicable requirements.

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.5 Screening. Crystal units to be delivered in accordance with this specification shall be subjected to and passed all screening tests of table I for the class specified, in the order shown, prior to performance of any quality conformance inspection. Unless otherwise specified (see 3.1), the percentage defective allowed (PDA) for lot acceptance shall be 10 percent for class B and 5 percent for class S.

4.5.1 Sample. A sample shall consist of all crystal units (100 percent screening).

4.5.2 Failures. Crystal units which fail any test criteria in the screening sequence shall be removed from the lot at the time of observation or immediately at the conclusion of the test in which the failure was observed. Once rejected and verified as a failure, rework may be performed.

4.5.3 Lots resubmitted for screening. Lots may be resubmitted for screening one time only and may be resubmitted only when the observed percentage of defectives does not exceed twice the specified percent defective allowed. For acceptance, reworked crystal units shall be subjected to and pass the complete screening sequence for the applicable device class.

TABLE I. Screening (100 percent).

Test inspection	Class S Method paragraph	Class B Method paragraph
Thermal shock	MIL-STD-202, method 107, condition C1.	Not applicable
Resistance (R_1 room temperature)	4.9.5.1, 20°C ±5°C	4.9.5.1, 20°C ±5°C
Radiographic (when specified)	MIL-STD-202, method 209	Not applicable

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4.6 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, on crystal units not covered by the specification sheets. The inspections shall be performed at a location acceptable to the government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the government to other contracts.

4.6.1 Sample size. Fifteen sample units of a single frequency and of a single type shall be subjected to first article inspection as specified in 4.6.2. For first article, a spread of frequencies of ±10 percent shall be considered as a single frequency.

4.6.2 Inspection routine. First article inspection shall consist of all inspections in table II in the order shown. One sample crystal unit is to remain at the contractor's plant and is to be available as a standard for comparative purposes.

4.6.2.1 First article data. The first article test plan and test report(s) shall be specified in the contract or purchase order (see 6.2).

4.6.3 Failures. If one or more sample crystal units fail to meet any of the first article requirements, the contractor shall immediately make the remedial changes. The contractor, at no additional cost to the government, shall be required to submit additional first article samples for reinspection. A description of the corrective action taken or to be taken shall be included in the first article inspection reports.

4.7 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the government (see 3.4 and 6.4) on sample units produced with procedures and equipment normally used in production.

4.7.1 Sample size. Forty-five sample units shall be subjected to qualification inspection as specified in 4.7.2.

4.7.2 Inspection routine. Qualification inspection shall consist of all inspections in table II in the order shown. One sample crystal unit is to remain at the contractor's plant and is to be available as a standard for comparative purposes.

4.7.3 Failures. Two failures in any one test shall be cause for refusal to grant qualification. Failure of one sample unit in one or more inspections shall be considered a single failure.

4.7.4 Retention of qualification.

4.7.4.1 Class B crystal units. To retain qualification for class B units, the contractor shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery, group A, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reinspected lots shall be identified and accounted for.

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b. The results of tests performed for periodic inspection, group B, including the number and mode of failures. The test report shall include results of all periodic inspection tests performed and completed during the 12-month period. If the test results indicate nonconformance with the specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during three consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit crystal units of the same type and range of frequencies to testing in accordance with the qualification inspection requirements.

4.7.4.2 Class S crystal units. To retain qualification for class S units, the manufacturer shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall include the following:

a. A list of all device types for which the manufacturer has qualification, showing the total number of lots completing quality conformance inspection during the reporting period.

b. Certification that the manufacturer still has the capabilities and facilities necessary to produce these items to the qualified tests.

c. The number of lots which failed lot-by-lot testing.

d. For failed lots the following information is required:

(1) Identification of the lots by inspection lot identification, date and lot code, part number, and date of completion of group B testing.

(2) The percent defective of the thermal shock and room temperature resistance screen.

(3) The results of groups A and B testing (i.e., pass or fail).

(4) For failed groups, the number and mode of all failures.

(5) All lots withdrawn shall be counted as failed lots in the report.

If the summary of test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

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In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements.

4.7.4.2.1 Records. Test records shall be in accordance with the format in MIL-STD-790. The report shall also include certification that the design and construction of the product was verified and found to be identical to the class S crystal unit for which qualification approval was valid. The manufacturer shall maintain group B samples for a minimum of 5 years.

TABLE II. Qualification and first article inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup I (all sample units)		
Solderability or lead attachment	3.7	4.9.3
Solderability (when applicable)	3.7.1	4.9.3.1
Ceramic package (when applicable)	3.7.2	4.9.3.2
Resistance to solvents (Four sample units)	3.8	4.9.4
Resistance to soldering heat (when applicable)	3.9	4.9.5
Visual (external) and mechanical inspection	3.1.3,3.5,3.6 3.33 and 3.34	4.9.2.1
Frequency and resistance	3.10	4.9.6
Resistance vs temperature (R vs T)	3.10.1	4.9.6.1
Frequency vs temperature (static)	3.10.2	4.9.6.1
Coupled modes (frequency-resistance amonalies)	3.10.3	4.9.6.2
Subgroup I (all sample units)		
Capacitance	3.11	4.9.7
Capacitance, shunt (when specified)	3.11.1	4.9.7.1
Capacitance, motional (when specified)	3.11.2	4.9.7.2
Quality factor ("Q") (when specified)	3.12	4.9.8
Unwanted modes	3.13	4.9.9
Insulation resistance	3.14	4.9.10
Accelerated aging	3.15	4.9.11
Drive sensitivity (of frequency and resistance)	3.16	4.9.12
Condensibles	3.17	4.9.13
Subgroup II (1/3 of sample units)		
Shock	3.18	4.9.14
Vibration, acceleration, and acoustical noise	3.19	4.9.15
Frequency and resistance offset (vibration) (when specified)	3.19.1	4.9.15.1
Acceleration sensitivity (vibration) (when specified)	3.19.2	4.9.15.2
Frequency and resistance offset (steady state acceleration) (when specified)	3.19.3	4.9.15.3
Acceleration sensitivity (steady state) (when specified)	3.19.4	4.9.15.4
Acoustical noise (when specified)	3.19.5	4.9.15.5
Thermal frequency repeatability (when specified)	3.20	4.9.16
Thermal frequency hysteresis	3.20.1	4.9.16.2

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TABLE II. Qualification and first article inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph
Subgroup III (1/3 of sample units)		
Particle impact noise detection (PIND) (when specified)	3.21	4.9.17
Moisture resistance	3.22	4.9.18
Salt spray	3.23	4.9.19
Terminal strength (Two sample units)	3.24	4.9.20
Terminal pull (when applicable)	3.24.1	4.9.20.1
Terminal bend (when applicable)	3.24.2	4.9.20.2
Wire-lead bend (when applicable)	3.24.3	4.9.20.3
Wire-lead twist (when applicable)	3.24.4	4.9.20.4
Visual (internal) 1/	3.1,3.5,3.6, and 3.33	4.9.2.2
Bond strength (when specified) 1/	3.25	4.9.21
Subgroup IV (1/3 of sample units)		
Thermal shock	3.26	4.9.22
Thermal time constant (when specified)	3.27	4.9.23
Frequency overshoot	3.27.1	4.9.23.1
Subgroup V (all sample units from subgroups II and IV)		
Barometric Pressure	3.28	4.9.24
Seal	3.29	4.9.25
Short term stability (when specified)	3.30	4.9.26
Aging	3.31	4.9.27
Radiation hardness	3.32	4.9.28
Total dose (when specified)	3.32.1	4.9.28.1
Dose rate (when specified)	3.32.2	4.9.28.2
Neutrons (when specified)	3.32.3	4.9.28.3
Accumulated time error (when specified)	3.32.4	4.9.28.4
Subgroup VI (Two unsealed units)		
Visual (internal)	3.1,3.5,3.6, and 3.32	4.9.2.2
Bond strength	3.25	4.9.21

1/ Six sample units other than those from terminal strength.

4.8 Quality conformance inspection.

4.8.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.8.1.1 Testing after storage. When a crystal unit is stored for 30 days or longer after group A inspection and before shipment, it shall be subjected to the frequency and resistance test (see 4.9.5).

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4.8.1.2 Inspection lot. An inspection lot shall consist of all crystal units of identical type and one frequency or assorted frequencies within a range covered by that type produced under essentially the same conditions, and offered for inspection at one time.

4.8.1.3 Group A inspection. Group A inspection shall consist of the inspections specified in table III in the order shown.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method
Subgroup 1		
Visual (external) and mechanical inspection	3.1,3.5,3.6, 3.32 and 3.33	4.9.2.1
Shock	3.18	4.9.14
Frequency	3.10	4.9.6
Resistance vs temperature (R vs T)	3.10.1	4.9.6.1
Frequency vs temperature (static)	3.10.2	4.9.6.1
Coupled modes (frequency-resistance anomalies)	3.10.3	4.9.6.2
Condensibles	3.17	4.9.13
Unwanted modes	3.13	4.9.9
Capacitance	3.12	4.9.7
Capacitance, shunt (when specified)	3.12.1	4.9.7.1
Capacitance, motional (when specified)	3.12.2	4.9.7.2
Quality factor ("Q") (when specified)	3.12	4.9.8
Seal	3.29	4.9.25
Subgroup 2		
Accelerated aging	3.15	4.9.11
Drive sensitivity (of frequency and resistance)	3.16	4.9.12

4.8.1.3.1 Class S crystal units. Group A inspection shall be performed on 100 percent of the lot for class S crystal units.

4.8.1.3.1.1 Class S lot rejection criteria. If 10 percent or more of the units in a lot fail accelerated aging (see 3.14), drive sensitivity (see 3.15), or condensibles (see 3.16), that lot shall be rejected. Resubmission is prohibited.

4.8.1.3.2 Class B crystal units. The tests of subgroup 1 shall be performed on the same set of sample crystal units as selected in accordance with the sampling plan (see 4.8.1.3.2.1). The tests in subgroup 2 may be performed on a separate set of sample units similarly selected.

4.8.1.3.2.1 Sampling plan. Sample lots shall be in accordance with appendix E, table VIII. The sample shall be representative of the inspection lot, with respect to frequency distribution, and containing as far as practicable, proportions equivalent to those of the inspection lot. When inspection lots, as defined in 4.8.1.2, consist of less than 90 units, two or more such lots may be combined into grand lots whose size does not exceed 90 units.

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4.8.1.3.2.2 Lot rejection criteria. A lot shall be rejected for one or more defects for that sample size as specified in appendix E, table VIII. Submission shall be in accordance with 4.8.1.3.3.

4.8.1.3.3 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units. The lot shall then be sampled in accordance with appendix E, table VIII and no defects are allowed. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. If any defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.8.1.3.4 Disposition of sample units. Sample units which have passed all the group A inspection may be delivered if the lot is accepted and the sample units are still within the specified tolerance with respect to frequency and resistance (see 3.1).

4.8.2 Periodic inspection. Periodic inspection shall consist of group B inspection.

4.8.2.1 Class S. Delivery of class S product shall be delayed pending acceptance of group B inspection for the production lot.

4.8.2.2 Class B. Except where the results of these inspections show noncompliance with the applicable requirements, delivery of class B products, which have passed group A inspections, shall not be delayed pending the results of these periodic inspections.

4.8.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV in the order shown. Group B inspection shall be made on sample units which have passed group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Subgroup I		
Solderability or lead attachment 1/	3.7	4.9.3
Solderability (when applicable)	3.7.1	4.9.3.1
Ceramic package (when applicable)	3.7.2	4.9.3.2
Resistance to solvents (Four sample units)	3.8	4.9.4
Vibration, acceleration, and acoustical noise	3.19	4.9.15
Frequency and resistance offset (vibration) (when specified)	3.19.1	4.9.15.1
Acceleration sensitivity (vibration) (when specified)	3.19.2	4.9.15.2
Frequency and resistance offset (steady state acceleration) (when specified)	3.19.3	4.9.15.3
Acceleration sensitivity (steady state) (when specified)	3.19.4	4.9.15.4
Acoustical noise (when specified)	3.19.5	4.9.15.5
Thermal shock	3.26	4.9.22
Thermal time constant (when specified)	3.27	4.9.23
Frequency overshoot	3.27.1	4.9.23.1
Thermal frequency repeatability (when specified)	3.20	4.9.16
Thermal frequency hysteresis	3.20.1	4.9.16.2

See footnote at end of table.

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TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Subgroup I - Continued		
Resistance to soldering heat	3.9	4.9.5
Moisture resistance	3.22	4.9.18
Salt spray	3.23	4.9.19
Particle impact noise detection (PIND) (when specified)	3.21	4.9.17
Terminal strength (Two sample units)	3.24	4.9.20
Terminal pull (when applicable)	3.24.1	4.9.20.1
Terminal bend (when applicable)	3.24.2	4.9.20.2
Wire-lead bend (when applicable)	3.24.3	4.9.20.3
Wire-lead twist (when applicable)	3.24.4	4.9.20.4
Visual (internal) 1/	3.25	4.9.21
Subgroup 2		
Insulation resistance	3.14	4.9.10
Short term stability (when specified)	3.30	4.9.26
Aging	3.31	4.9.27
Radiation hardness	3.32	4.9.28
Total dose (when specified)	3.32.1	4.9.28.1
Total dose (when specified)	3.32.2	4.9.28.2
Neutrons (when specified)	3.32.3	4.9.28.3
Accumulated time error (when specified)	3.32.4	4.9.28.4

1/ Six sample units other than those from terminal strength.

4.8.3.1 Sampling plan.

4.8.3.1.1 Class B. Inspection shall be in accordance with appendix E, table IX. The lot or batch shall consist of all inspection lots of the same crystal type that have passed group A inspection during the time period not to exceed 12-months. The sample shall be representative of the lot with respect to frequency distribution, containing as far as practicable, proportions equivalent to those of the lot.

4.8.3.1.2 Class S. Inspection shall be in accordance with appendix E, table IX. The sample shall be representative of the lot with respect to frequency distribution, containing as far as practicable, proportions equivalent to those of the lot.

4.8.3.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order. The samples shall be maintained with relative data and traceability for not less than 5 years.

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4.8.3.3 Noncompliance.

4.8.3.3.1 Class B crystal units. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed, at the option of the qualifying activity). Group A inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.8.3.3.2 Class S crystal units. If an inspection lot fails to pass group B inspection, the inspection lot shall not be delivered on the contract or purchase order. The manufacturer shall notify the qualifying activity and cognizant inspection activity of the failure and take corrective action on the materials or processes, or both as warranted. A failure analysis (see 4.8.3.4) shall be performed on the failing product and forwarded to the qualifying activity. A copy of the results shall be maintained by the manufacturer. If three successive lots of any type of crystal unit or three successive lots of a single type of crystal fail quality conformance testing (groups A and B), the qualifying activity shall be notified within 96 hours and the qualifying activity, at its discretion, may remove the failing product from the qualified products list.

4.8.3.4 Failure analysis. If any of the sample units subjected to the group B tests fail during testing, a detailed failure analysis shall be conducted to establish the cause of failure and the corrective actions that would eliminate subsequent failures of a similar type. A failure is categorized as lot oriented if its occurrence is apparently related to an identified lot or lots. A failure is categorized as not lot oriented if its occurrence is random and it cannot be related to a specific lot or lots. Each failure is further identified as screenable or not screenable from the completed production items. If the failure analysis shows that the failure mechanism is screenable, the entire failed lot may be screened and the group B test in which the failure occurred shall be repeated. If a failure occurs during the second group B test, the entire production lot shall be rejected. If the failure mechanism is screenable, all prior and subsequent production lots that may contain the identified failure mechanism shall also be screened. Except as may be stated otherwise in the detailed requirements for the specific part type (see 3.6), if the failure mechanism is lot oriented and not screenable, all production lots that may contain the identified failure mechanism shall be rejected unless other disposition is directed by the contracting officer and qualifying activity.

4.8.4 Certification. The manufacturer shall certify with each order of parts, that the applicable quality conformance tests were successfully completed.

4.8.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.9 Methods of inspection.

4.9.1 *Test criteria*.

4.9.1.1 Reference plane and grounding requirements. Because of lead inductance, it is necessary to specify a reference plane at which the measurements are to be made. This plane is located at a distance of 2 mm (0.080 inch) from the place where the pins or wires emerge from the holder-base, unless otherwise specified (see 3.1). Metal enclosed crystal units shall be measured with the holder grounded. A grounded shielding box as specified in IEC-444 shall be provided for non-metallic enclosures.

4.9.1.2 Absolute measurements. When measurements are made to show conformance to a specification, the drive current shall be as specified (see 3.1) and the reference and ground plane shall be as specified in 4.9.1.1. The measurements shall be traceable to the contractor's standard test set (see 4.1.3.1).

4.9.1.3 Relative measurements. When frequency and resistance are measured before and after a test to determine the change which took place during a test, both measurements shall be made ideally with the crystal unit in thermal equilibrium at the same temperature and with the drive level, reference plane and ground configuration identical. If the temperature cannot be controlled to the degree necessary for the required precision, a correction may be applied using measured f versus T data. The temperature shall be controlled such that the correction is no larger than 30 percent of the allowed frequency change. The drive level and reference plane shall be reproduced to a degree consistent with the required precision.

4.9.2 Visual and mechanical inspection.

4.9.2.1 External. Crystal units shall be examined to verify that the external design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.5, 3.6, 3.33 and 3.34).

4.9.2.1.1 Glass parts. A strong light and 10-power magnification shall be used to examine the glass parts (see 3.6.4 and 3.33).

4.9.2.2 Internal. All crystal units, except glass-enclosed, shall be disassembled or if pre-seal inspection is specified in the contract (see 6.2), crystal units shall be inspected prior to sealing, and the interior shall be examined to verify that the internal design, construction and workmanship are in accordance with the applicable requirements (see 3.1, 3.5, 3.6, 3.6.1, 3.6.2, and 3.33). The crystal resonator shall be inspected with a 10-power magnification while illuminated with a 300 foot-candle light source having a grazing angle of approximately 20 degrees. For glass enclosed crystal units, internal inspection may be performed visually without disassembly. Measurement of wall thickness is not applicable for glass-enclosed crystal units.

4.9.3 Solderability or lead attachment (see 3.7).

4.9.3.1 Wire-lead terminals (see 3.7.1). Each wire-lead terminal shall be subjected to method 208 of MIL-STD-202.

4.9.3.2 Ceramic package (see 3.7.2).

4.9.3.2.1 Parallel-gap welding or thermocompression bonding. Gold or gold-plated wires or ribbons shall be attached using the specified method (see 3.1). Each wire or ribbon shall be subjected to test condition C, method 2011, of MIL-STD-883.

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4.9.3.2.2 Solderability. If soldering is specified, (see 3.1), a copper wire, 0.25 mm in diameter (number 30 B and S) shall be soldered to each terminal. Each wire shall be subjected to test condition A and test condition B (flexible leads), method 2004, of MIL-STD-883.

4.9.4 Resistance to solvents (see 3.8). The crystal units shall be tested in accordance with method 215 of MIL-STD-202.

4.9.5 Resistance to soldering heat (see 3.9). Crystal units shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Measurement before test: The frequency and equivalent resistance shall be measured as specified in 4.9.1.3 and 4.9.6.1.

b. Special preparation of the specimen: Leads shall be dipped in RMA flux in accordance with MIL-F-14256, and then dipped into solder, both for 5+1/2 seconds. The bath shall be maintained at 260°C +5°C. The parts shall be immersed to within .075+.025 inch of the body.

c. Test condition C (260°C +5°C, 10+2 seconds: Board top side shall reach 200°C mininum). A board with a maximum area of 9 square inches shall be used, and the leads shall not be cut. Crystal units shall be mounted on the circuit board in accordance with MIL-STD-275 mounting practices.

d. Post test conditioning: The crystal units shall be cooled to room temperature and then inserted in a vapor degreasing apparatus containing boiling 1-1-1 trichloroethane for 1-1/2 + 1/2, -0 minutes. The part shall then be cleaned with isopropyl alcohol.

e. Measurements after test: After completion of the cleaning process and following a minimum 3 hour cooling period, the frequency and equivalent resistance shall be measured as specified in 4.9.1.2 and 4.9.9.

f. Examination after test: Crystal units shall be examined for evidence of mechanical damage.

4.9.6 Frequency and resistance (see 3.10)

4.9.6.1 Frequency and resistance versus temperature (static temperature run). The crystal unit shall be measured with the specified series load capacitor (see 3.1) or with an equivalent phase offset described in 4.9.7.2.1 at the specified drive current. The load capacitor shall have a temperature coefficient commensurate with the required accuracy. The temperature shall be varied at the rate specified (or slower) over the range specified. Frequency and resistance data shall be recorded at least as often as specified (see 3.1). In the region of the turnover temperature or inflection temperature, the rate of change of temperature and the data recording interval shall be adjusted as specified. The data shall be analyzed statistically or graphically to determine the required frequencies, temperatures and slopes (see 3.1). Unless otherwise specified, the maximum rate of change of temperature shall not exceed 10°C/minute.

4.9.6.2 Coupled modes (see 3.10.3). The frequency versus temperature data shall be analyzed by determining the best fit to the specified order polynomial (see 4.9.7.2.1). All frequency residuals (see 4.9.7.2.1) shall be less than the value specified. If specified, the resistance versus temperature data shall be analyzed by determining the maximum slope (averaged over a 5°C temperature interval) and the maximum resistance in the temperature range specified (see 3.1).

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4.9.6.2.1 Polynomial and residuals.

$$f \text{ versus } T: f(T) = f(0^{\circ}C)\,(1+AT+BT^2+CT^3+DT^4+ET^5+FT^6)$$

Frequency residual = f(measured at T) - f(calculated from polynomial at T)

4.9.7 Capacitance.

4.9.7.1 Capacitance, shunt (see 3.11.1). Crystal units shall be tested in accordance with method 305 of MIL-STD-202. The capacitance shall be measured from terminal to terminal with the crystal holder grounded at a frequency at which the resonator shows no resonance response.

4.9.7.2 Capacitance, motional (see 3.11.2). The motional capacitance, C_1, is defined in accordance with IEC-302. The load resonant frequency (f_L) is measured using two load capacitances connected successively in series with the crystal unit. The results obtained with two load capacitors C_{L1} and C_{L2} can be combined so that:

$$C_1 = \frac{2\Delta C_L \Delta f_1 \Delta f_2}{f_r \Delta f}$$

where

ΔC_L = $C_{L2} - C_{L1}$
Δf = $f_{L1} - f_{L2}$
Δf_1 = $f_{L1} - f_r$
Δf_2 = $f_{L2} - f_r$
f_r = Resonance frequency

It is necessary to maintain the current through the crystal unit constant to within 10 percent and the frequencies must be corrected for temperature variations during the test, if necessary. The two C_L's shall be chosen such that Δf, Δf_1 and f_2 are large enough with respect to the frequency resolution of the measurement system to deliver the required accuracy.

4.9.7.2.1 Phase offset method. An alternate method for determining the value of C_1 is the phase offset method. This method is useful in a transmission or reflection type vector impedance system. The exact implementation is hardware dependent. In principle, the frequency is offset an amount, Δf, from resonance and the phase shift is measured. The admittance is calculated and the shunt capacitance of the fixture and the crystal unit is subtracted. The same measurement is performed at $-\Delta f$ from resonance. Using the motional resistance, R_1, the motional reactance at $+\Delta f$ is determined. The motional inductance is calculated from:

$$L_1 = (1/8\ \Delta f)\,(X_1(\Delta f) - X_1(-\Delta f))$$

C_1 is calculated from

$$C_1 = \frac{1}{(2\ f_r)^2 L_1}$$

4.9.8 Quality factor (see 3.12). The quality factor, Q, as defined by

$$Q = \frac{1}{2\ f_r\ R_1 C_1}$$

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shall be determined using the largest value of R_1 measured in the temperature range specified (see 3.1). R_1 shall be calculated from R_L using the following equation

$$R_1 = \frac{R_L}{(1 + C_o/C_L)^2}$$

4.9.9 Unwanted modes (see 3.13). The resistance of any unwanted mode shall be measured in the same manner as the desired mode (see 4.9.1.2).

4.9.10 Insulation resistance (see 3.14). Crystal units shall be measured in accordance with method 302 of MIL-STD-202, using a test potential of 50 to 55 volts. This measurement shall be made from terminal to terminal and from each terminal to the holder case, if the holder is metal.

4.9.11 Accelerated aging (see 3.15). Crystal units shall be measured for frequency and resistance (see 4.9.1.3) and then conditioned in an oven at 125°C for 160 hours ±1 hour. The crystal units shall then be removed from the oven and measured for frequency and resistance in accordance with 4.9.1.3.

4.9.12 Drive sensitivity (of frequency and resistance (see 3.16)). The crystal units shall be conditioned at a temperature of at least 125°C for a duration of at least 1 hour. After conditioning, the units shall be stored at room temperature for 3-6 hours. Measurement of the resonant frequency and resistance shall be made at no fewer than 4 drive levels between 5 microamperes or less and the rated drive level. The levels shall be chosen at nominally equal intervals on a logarithmic scale. The sequence of measurements shall be from lowest drive current to highest drive current. The crystal unit shall not have been operated from the beginning of the conditioning period to the start of the frequency and resistance measurements at the lowest drive level. A drive level 10 dBm above the minium is allowed to find the response

4.9.13 Condensibles (see 3.17).

4.9.13.1 Metal or ceramic package. The crystal units shall be subjected to a temperature of 125°C for at least 10 minutes. The crystal units shall then be rapidly cooled by immersion in a liquid at -55°C ±5°C for a minimum of 5 minutes. Subsequent to exposure to -55°C, the crystal units shall be heated at a rate of at least 10°C/minute, while the resistance is continuously monitored, to a temperature of 75°C ±5°C.

4.9.13.2 Glass package. The crystal units shall be subjected to a temperature of 125°C for at least 10 minutes. The crystal units shall then be cooled to -55°C ±5°C for a minimum of 5 minutes. Subsequent to exposure to -55°C, the crystal units shall be heated at a rate of at least 10°C/minute, while the resistance is continuously monitored to a temperature of 75°C ±5°C.

4.9.14 Shock (see 3.18). Crystal units shall be tested in accordance with method 213 of MIL-STD-202. The following details apply:

4.9.14.1 Test conditions. The test condition shall be as specified in 3.1.

4.9.14.2 Measurements. Frequency and resistance measurements shall be made before and after the test in accordance with 4.9.1.3.

4.9.15 Vibration, acceleration, and acoustical noise (see 3.19).

4.9.15.1 Frequency and resistance offset (vibration) (see 3.19.1). Crystal units shall be tested in accordance with MIL-STD-202, method 204 or method 214, as specified (see 3.1). The following details apply:

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4.9.15.1.1 Test conditions. The test condition for the specified method shall be as specified (see 3.1).

4.9.15.1.2 Duration. The duration shall be three hours or as specified (see 3.1).

4.9.15.1.3 Direction of motion. Specimens shall be rigidly mounted to the platen of a vibration machine so that the applied force shall be, unless otherwise specified (see 3.1), along one of three mutually perpendicular directions for each of three approximately equal sized groups (i.e., one-third of the units are to be vibrated along one direction, one-third of the units along a perpendicular direction and the remaining one-third vibrated along the direction mutually perpendicular to the previous two directions).

4.9.15.1.4 Measurements. The frequency and resistance shall be measured in accordance with 4.9.1.3.

4.9.15.2 Acceleration sensitivity (vibration) (see 3.19.2). The crystal unit, operating in an appropriate oscillator or other measurement system, shall be rigidly mounted to the platform of a vibration machine. A sinusoidal vibration shall be applied to the vibration machine at a level not to exceed 20g or the value specified (see 3.1). The frequency of vibration, fv, shall be varied over the frequency range specified (see 3.1) in a logarithmic progression of not fewer than 7 frequencies per decade. At each vibration frequency the relative sideband intensity, i.e., the ratio of the power at the unmodulated signal frequency, shall be determined. The acceleration sensitivity at each vibration frequency shall be determined from the following:

$$\gamma_i = \frac{2 f_v 10^{(I_i(f_v)/20)}}{a_i V_o}$$

where

γ_i = Acceleration sensitivity along axis i (i = 1, 2, or 3)
f_v = Vibration frequency
a_i = Acceleration frequency
V_o = Carrier frequency
$I_i(f_v)$ = Relative sideband intensity (power ratio) in dB when the vibration at frequency f_v is along the i axis.

The acceleration sensitivity for each axis is the maximum value measured for any vibration frequency in the range specified (see 3.1).

The above procedure is repeated for two other axes such that the three axes are mutually perpendicular. The acceleration sensitivity of the crystal unit shall be determined from the following:

$$|\vec{\Gamma}| = \sqrt{(\gamma_1)^2 + (\gamma_2)^2 + (\gamma_3)^2}$$

4.9.15.2.1 Mechanical resonance. A mechanical resonance is present if there is a peak in the acceleration sensitivity versus vibration frequency curve.

4.9.15.3 Frequency and resistance offset (steady state acceleration) (see 3.19.3). A steady state acceleration shall be applied at the specified level for the specified duration (see 3.1). Each inspection lot shall be divided such that one-third of the units are being accelerated in one of three mutually perpendicular directions. The frequency and resistance shall be measured before and after exposure in accordance with 4.9.1.3.

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4.9.15.4 Acceleration sensitivity (steady state) (see 3.19.4). The frequency shall be monitored in a suitable oscillator or other test set. The acceleration shall be produced using a centrifuge or similar machine. Measurements shall be made for at least 5 equally spaced acceleration levels between 20 percent of the maximum and the maximum specified. The frequency change per unit g of acceleration shall be determined graphically for all acceleration levels up to the maximum specified (see 3.1).

4.9.15.5 Acoustical noise (see 3.19.5). Crystal units shall be tested in accordance with procedure I, method 515, MIL-STD-810. The single sideband ratio of the power in a 1 Hz bandwidth to the power in the carrier shall be determined for the frequency range of 10 Hz to 10,000 Hz.

4.9.16 Thermal frequency repeatability (see 3.20). The crystal unit shall be subjected to the temperature cycle specified in 4.9.16.1. The frequency at each turnover temperature shall be determined using the method of least squares from the data obtained at the five temperatures immediately above and the five temperatures immediately below the turnover temperatures. The difference between the lower turnover frequency and the upper turnover frequency is defined as df.

4.9.16.1 Temperature cycle (See figures 1 and 2).

a. Cool from room temperature to within ±2°C of the lowest temperature specified (see 3.1) at a rate not to exceed 1°C/min. This temperature is defined as T_L.

b. Hold at T_L ±2°C for 2 hours ±15 minutes.

c. Raise temperature at a rate not to exceed 1°C/minute.

d. In the region of the turnover temperature, establish thermal equilibrium at least 10 temperatures spaced 1°C ±0.2°C apart, at least 5 of the temperatures above the turnover temperature and at least 5 of the temperatures below the turnover temperature.

e. Measure the frequency and temperature at each equilibrium point of step d, with a resolution commensurate with the required accuracy.

f. If there is another turnover temperature below the highest temperature specified (see 3.1), T_H, go to step C and continue, otherwise go to step g.

g. Raise the temperature to T_H ±2°C at a rate not to exceed 1°C/minute.

h. Hold at T_H ±2°C for 2 hours ±15 minutes.

i. Lower the temperature at a rate not to exceed 1°C/minute.

j. Same as step d.

k. Same as step e.

l. If there is another turnover temperature above T_L, go to step i and continue, otherwise go to step m.

m. Lower the temperature to T_L ±2°C at a rate not to exceed 1°C/minute.

n. Repeat b-m the specified number of times (see 3.1).

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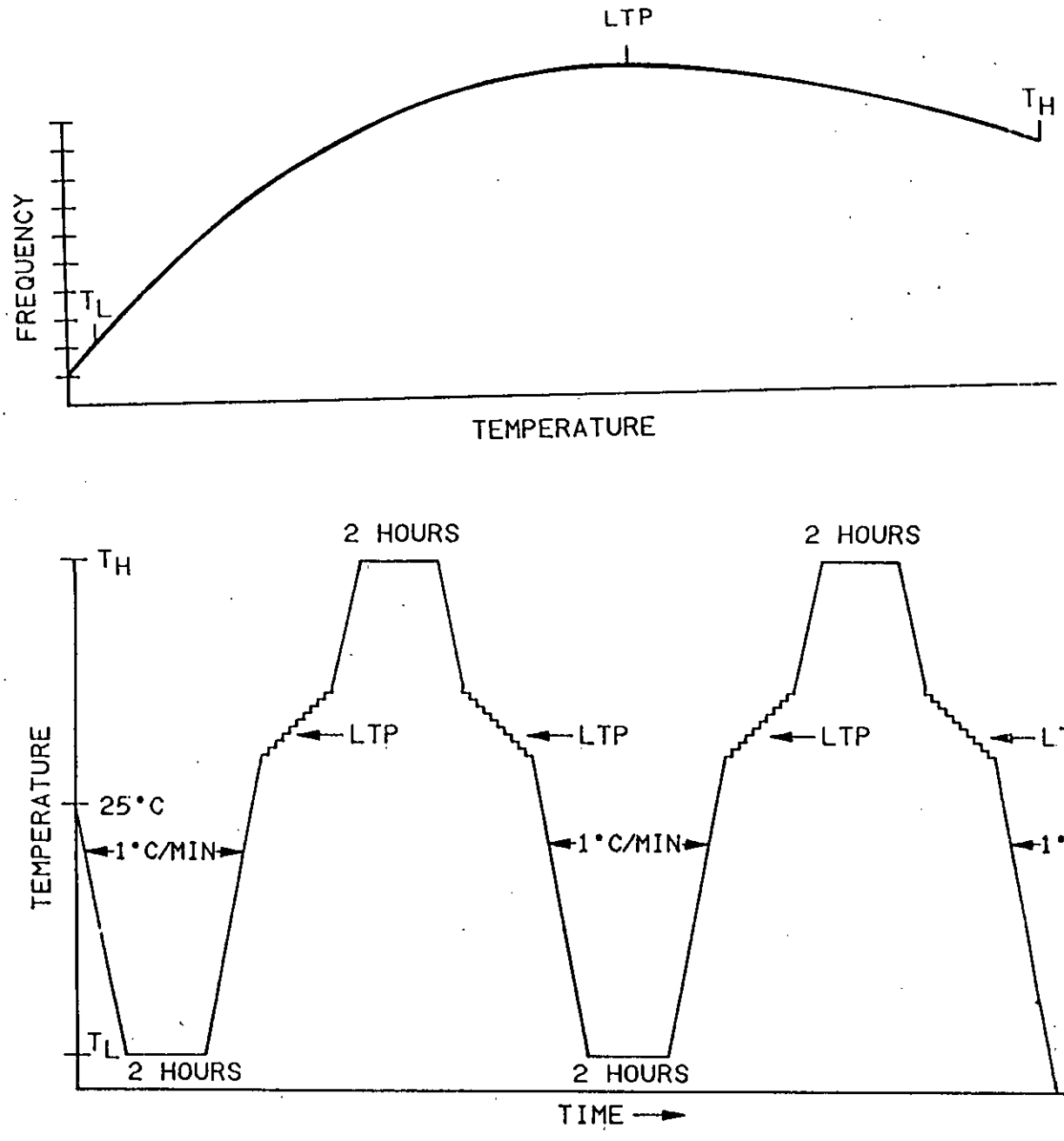
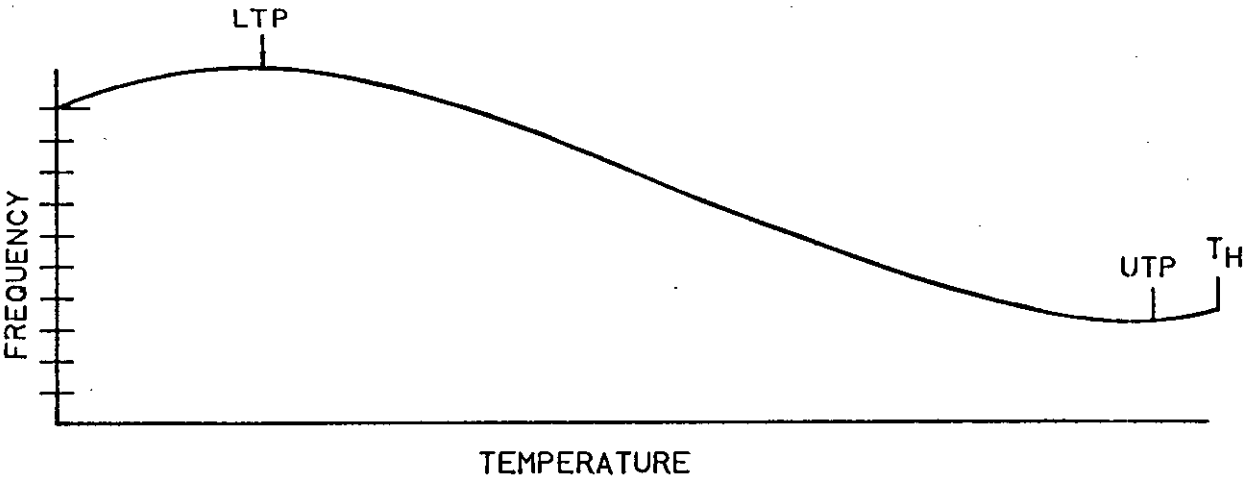


FIGURE 1. Thermal frequency repeatibility (one-turnover thermal cycle).

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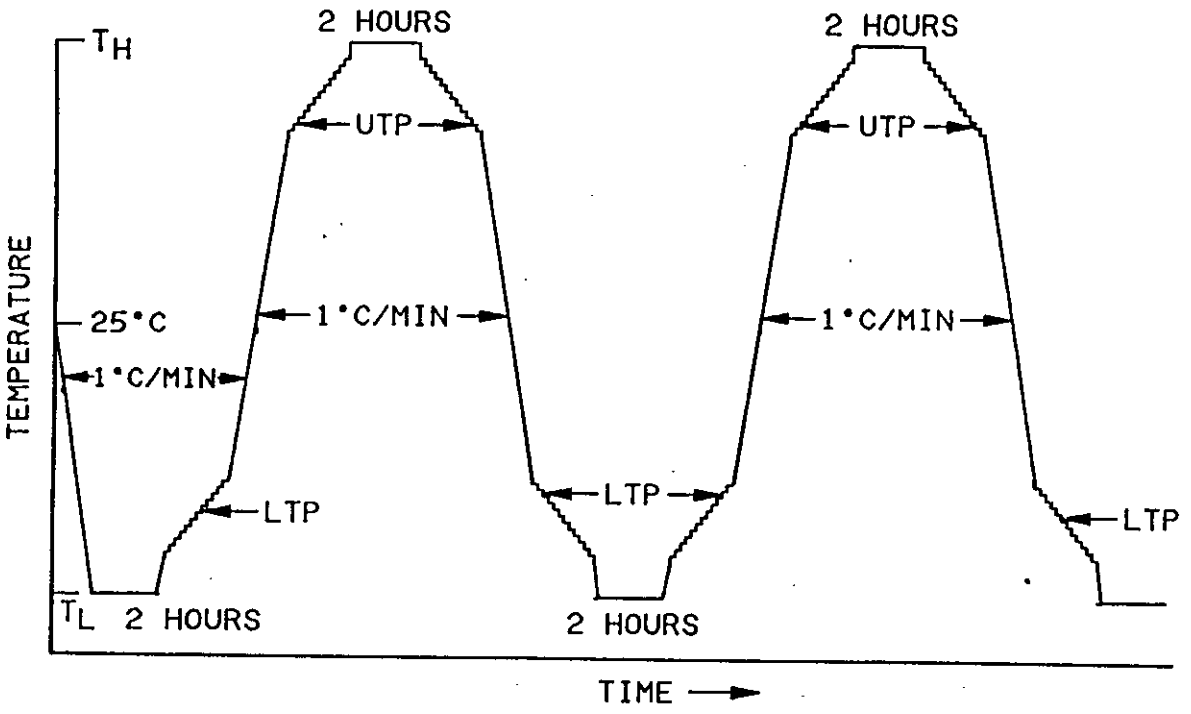


FIGURE 2. Thermal frequency repeatibility (two-turnover thermal cycle).

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4.9.16.2 Thermal frequency hysteresis (see 3.20.1).

a. Segment 1: The temperature of the crystal unit shall be stepped in increments, not to exceed 5°C, starting from the specified lower temperature limit, T_L, up to the upper temperature limit, T_H. At each step, after thermal equilibrium has been reached, temperature and frequency measurements shall be taken.

b. Segment 2: Step the temperature from T_H to T_L. Take temperature and frequency measurements as in segment 1.

c. Segment 3: Same as segment 1

d. Segment 4: Same as segment 2

e. Using a least square fit, determine the coefficients to a sixth order polynomial for segment 1.

f. Calculate four sets of residuals using the data from each segment and the polynomials determined in e above.

g. The thermal frequency hysteresis is 1/2 of the maximum difference among the values obtained by correcting the residuals obtained in f by straight line segments.

4.9.17 Particle impact noise detection (PIND) (see 3.21). Crystal units shall be tested as specified in method 2020 of MIL-STD-883, test condition A, for one cycle. Pre-shock level and co-shock level shall be specified in other than that specified in MIL-STD-883.

4.9.18 Moisture resistance (see 3.22). Crystal units shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Normal means, except during measurement.
b. Initial measurements: see 4.9.1.3.
c. Subcycle: Step 7B, the vibration subcycle, shall be omitted.
d. Polarization and loading voltage: Not applicable.
e. Final measurement: see 4.9.1.3.
f. Visual examination: After the test, the unit shall be visually examined for evidence of excessive corrosion.

4.9.19 Salt spray (see 3.23). Crystal units shall be tested in accordance with method 101, test condition B, of MIL-STD-202. After this test, the unit shall be visually examined for evidence of excessive corrosion.

4.9.20 Terminal strength (see 3.24).

4.9.20.1 Terminal pull (see 3.24.1). Crystal units shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

a. Test condition: A
b. Applied force: 4 pounds for pin terminals and 2 pounds for wire terminals, applied to each terminal separately.

4.9.20.2 Wire-lead bend (applicable to crystal units with wire-lead terminals) (see 3.24.2). Crystal units shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply.

a. Test condition: C
b. Applied load: 1 pound

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4.9.20.3 Wire-lead twist (applicable to crystal units with wire-lead terminals) (see 3.24.3). Crystal units shall be tested in accordance with method 211, test condition D, of MIL-STD-202.

4.9.21 Bond strength (see 3.25). The strength of the bond between each wire, ribbon or other support structure and the quartz resonator shall be tested in shear for thickness-shear type resonators and in tension for other type resonators, by applying a force along the support member until the specified bond strength is reached (see 3.1). The full force shall be applied for not more than 1 minute. Breakage of the quartz resonator during the test shall not be considered as a bond strength failure.

4.9.22 Thermal shock (see 3.26). Crystal units shall be tested in accordance with MIL-STD-202, method 107. The test condition shall be C, except for first article or qualification when the test condition shall be C1.

4.9.23 Thermal time constant (see 3.27). The crystal unit, in thermal equilibrium at 0°C ±1°C and operating in a suitable test system shall be rapidly (less than 0.5 seconds) immersed in a suitable liquid bath held at 100°C ±1°C and held for at least three minutes. The frequency of the crystal unit shall be recorded every 0.1 second starting at least 10 seconds before insertion into the 100°C bath and ending after the three minute soak. The time required for the frequency to change to its static value (see 3.9.2) at 63°C ±1°C shall be recorded as the thermal time constant. If the frequency crosses the 63°C value more than once, the longest time shall determine the time constant.

4.9.23.1 Frequency overshoot (see 3.27.1). The frequency overshoot shall be defined as:

$$\text{frequency overshoot} = (f_d - f_s)/f(100^{\circ}C)$$

using figures 3 and 4 as a guide and where

f_d = f_t (dynamic) - f(100°C) (dynamic)

f_s = f_t (static) - (f100°C) (static)

and

f_t = turnover frequency

f (100°C) = frequency at 100°C

The static values are obtained from the data taken in support of 4.9.6.1. The dynamic values are obtained from the data taken in support of 4.9.23.

4.9.24 Barometric pressure (reduced) (when specified, see 3.28). Crystal units shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Mounting: Normal means

b. Test condition letter: Test condition B

c. Tests during subjection to reduced pressure: None

d. Tests after subjection to reduced pressure: Visual examination, the unit shall be visually examined for evidence of any seal rupture.

4.9.25 Seal (see 3.29). The crystal units shall be subjected to the procedures specified in 4.9.23.1 and 4.9.23.2. Fine leak check shall precede gross leak check.

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4.9.25.1 Fine leak check. Crystal units shall be tested in accordance with method 112 of MIL-STD-202. The following details apply:

a. Test condition: C
b. Procedure - III: Without tracer gas
IV : With tracer gas
c. Leakage rate sensitivity: Ten percent of the specified leak rate

4.9.25.2 Gross leak check. Crystal units shall be tested in accordance with method 112 of MIL-STD-202, test condition A, B, D, or E.

4.9.26 Short term stability (see 3.30). The short term stability test shall be performed in a suitable test system during the last 7 days of the aging cycle (see 3.29). The short term stability shall be defined as the Allan variance using 100 samples (see NIST Technical Note 1337 - Characteristic of Clocks and Oscillators.)

4.9.27 Aging (see 3.31). Crystal units shall be maintained at the specified aging temperature (see 3.1) for a continuous period of 30 days. Resistance measurements shall be made prior to inserting the crystal units into the oven and after removal from the oven upon completion of the test. After insertion in the oven, the crystal units shall be stabilized for two days at the aging temperature prior to beginning the data acquisition cycle. The time origin for data analysis shall be the beginning of the stabilization period. The frequency of each unit shall be measured immediately after the stabilization period. The frequency of each unit shall be measured a minimum of 4 times per week for 28 days at intervals of at least 20 hours. The data obtained shall be fit using the method of least squares to the function:

$$f(t) = A(\ln(Bt+1))+f_0$$

where f(t) is the frequency of the crystal unit t days after the start of the aging cycle, and A, B, and f_0 are constants to be determined from the least square fit. The calculated frequency f_0 is for the beginning of the aging cycle.

If the aging trend is not monotonic, the measurement period shall be extended to 40 days after the extremum in the aging trend. For the total aging period of n days, the data for the last 28 days shall be fit to the above function.

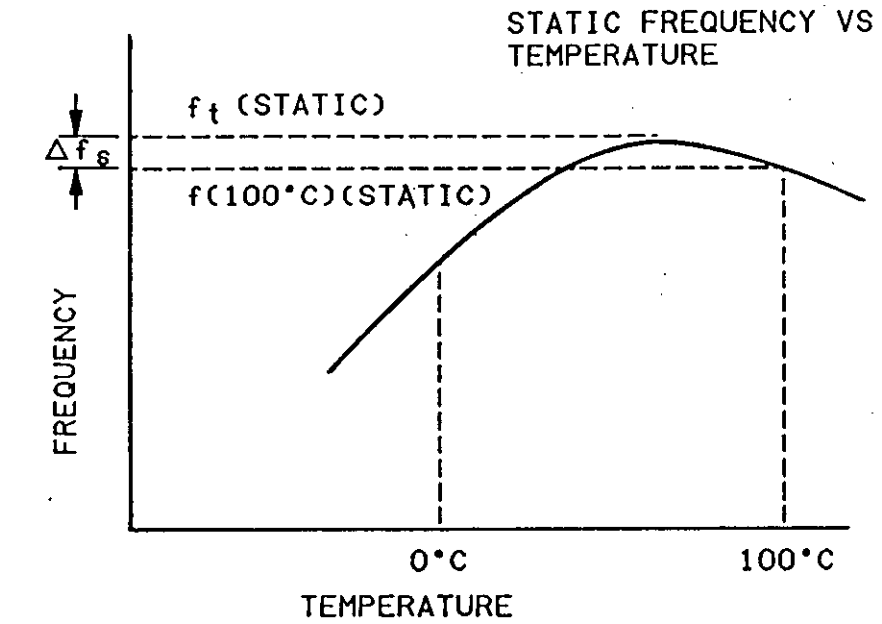
The aging rate per day at day n and the total frequency change from day n-30 to day n shall be determined from the equation using the above determined constants. The maximum frequency shift between any two successive data points shall be determined from the original data.

The square root least square fit variance of the data from the function shall not exceed 10 percent of the total aging change allowed during the test period.

4.9.28 Radiation hardness (see 3.32).

4.9.28.1 Total dose (see 3.32.1). Total dose radiation testing shall be performed in accordance with method 1019 of MIL-STD-883. The crystal units shall be subjected to the specified total dose (see 3.1). For total dose levels below 1 mrad, the frequency shall be measured during the irradiation. The temperature of the crystal units shall be maintained at a turnover temperature. For total dose levels above 1 mrad, the frequency shall be measured within four days following irradiation in accordance with 4.9.1.3. The resistance shall be measured in accordance with 4.9.1.3 within four days following irradiation.

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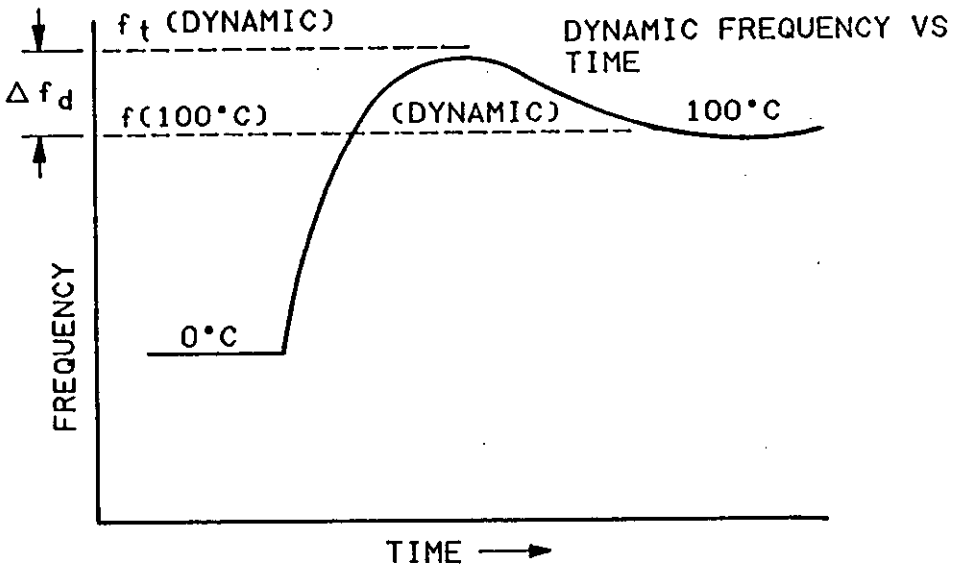
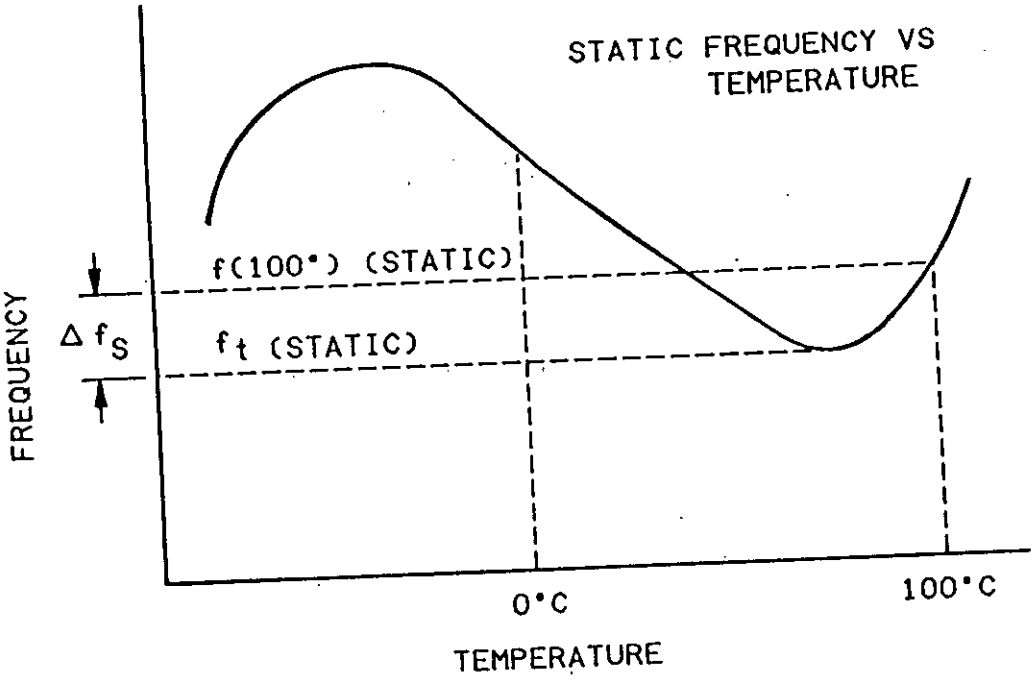


FIGURE 3. Frequency overshoot (upper turnover temperature between 0°C and 100°C.

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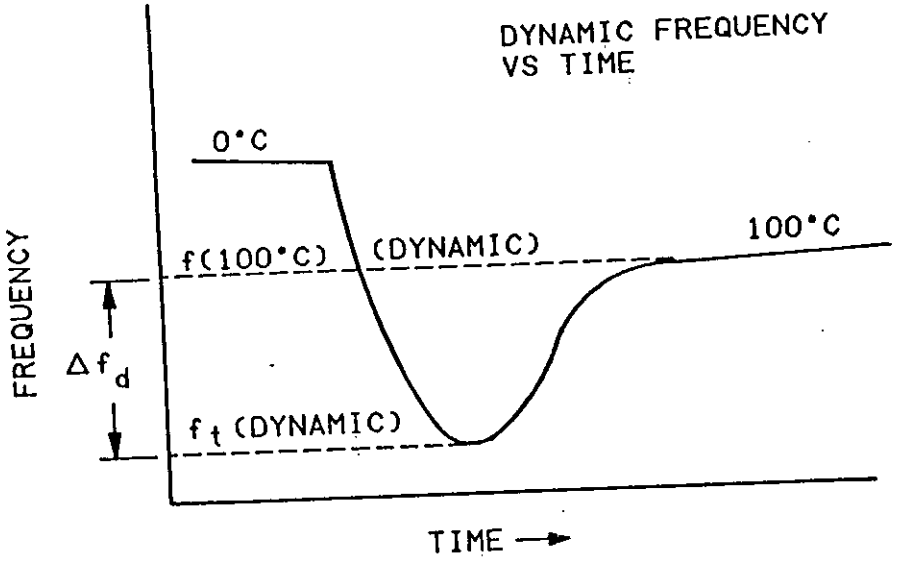


FIGURE 4. Frequency overshoot (lower turnover temperature between 0°C and 100°C; upper turnover temperature above 100°C.

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4.9.28.2 Dose rate (see 3.32.2). The crystal units shall be subjected to the specified radiation pulse. The frequency shall be measured during the irradiation and for the specified time subsequent to irradiation. The temperature of the crystal units shall be maintained at a turnover temperature during the test.

4.9.28.3 Neutrons (see 3.32.3). The crystal units shall be subjected to the specified neutron fluence. The frequency shall be measured during irradiation. The temperature of the crystal unit shall be maintained at a turnover temperature during the test. The resistance shall be measured in accordance with 4.9.1.3. The frequency change due to any incidental non-neutron irradiation shall be properly accounted for.

4.9.28.4 Accumulated time error (see 3.32.4). The crystal unit shall be subjected to the specified total dose at the specified dose rate. The time integral of the radiation-induced frequency offset, for the specified time interval (see 3.1), shall be determined.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Crystal units shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Crystal units shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packs. Crystal units shall be unit packed one each in accordance with method III of MIL-P-116 insuring compliance with the applicable requirements of that specification.

5.1.1.5 Intermediate packs. Crystal units, unit packed as specified in 5.1.1.4, shall be placed in intermediate containers conforming to variety 2 of either PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape, and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packs, not to exceed 100 unit packs. No intermediate packs are required when the total quantity shipped to a single destination will result in only one intermediate pack per shipping container.

5.1.2 Level B. The requirements for level B shall be as specified for level A except that any variety of the intermediate containers specified may be used.

5.1.3 Level C. The level C preservation of crystal units shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Crystal units, preserved as specified in 5.1, shall be packed in wood boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. The requirements for level B packing shall be used when the total quantity of a stock numbered crystal unit for a single destination does not exceed a packed volume of one cubic foot.

5.2.2 Level B. Crystal units, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. The requirements for box closure, waterproofing and reinforcing shall be in accordance with method V of the PPP-B-636 appendix.

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5.2.3 Level C. Crystal units, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or part number, as applicable (including the CAGE), shall be marked on all unit and supplementary packs in accordance with the identification marking provisions of MIL-STD-129.

5.4 General. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.8.5.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Crystal units are intended to be used as specified (see 3.1).

6.1.1 Packaging requirements. The preservation, packing and marking specified herein are intended for direct shipments to the Government. However, at the option of the contractor or when so specified, the packaging provisions herein are also applicable for the preparation of crystal units for shipment from the parts contractor to the original equipment manufacturer.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of the applicable specification sheet and the complete Part or Identifying Number (PIN) (see 1.2.1 and 3.1).
c. Frequency or frequencies required (see 1.2.2).
d. If a contract calls for modification of any requirements of the specification sheet, the PIN is not to be used to identify the crystal unit. Any modifications shall follow the guidelines set forth in appendix B.
e. Any additional marking required on the crystal unit (see 3.32).
f. When materials certification is not required (see 3.5).
g. Where first article and quality conformance inspections are to be performed (see 4.1).
h. If unsealed units are required for preseal inspection (see 4.9.2.2).
i. Levels of preservation and packing required (see 5.1 and 5.2).
j. If special or additional identification marking is required (see 5.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be on crystal units not covered by specification sheets or units with a specification sheet but no QPL. The first article should consist of the same sample units as qualification. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts for the products covered by this specification. The activity responsible for the Qualified Products List is US Army Laboratory Command, SLCET-R-S, Fort Monmouth, NJ 07703-5302; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, OH 45444.

6.5 Definitions. The following definitions apply:

6.5.1 Crystal unit. A crystal unit is an assembly that consists of a quartz resonator (quartz blank plus electrodes) suitably mounted within a crystal holder.

6.5.2 Crystal holder. A crystal holder is the sealed enclosure in which a quartz resonator is mounted; it includes a cover or covers, a base or frame with suitably insulated pins or terminals and some means for closure. Designation of a holder does not designate the mounting structure for the resonator.

6.5.3 Frequency range. The frequency range of a crystal unit type is the range from minimum to maximum in which any frequency within this range may be specified.

6.5.4 Nominal frequency ($f_{nominal}$). The nominal frequency is the frequency at which the crystal unit is designed to operate under the conditions specified by the specification sheet.

6.5.4.1 Resonance frequency (f_r). The lower of the two frequencies of the crystal unit alone, under specified conditions, at which the electrical impedance of the crystal unit is resistive.

6.5.4.2 Antiresonance frequency (f_a). The higher of the two frequencies of a crystal unit alone, under specified conditions, at which the electrical impedance of the crystal unit is resistive. This frequency is not normally used in frequency control applications.

6.5.4.3 Load resonance frequency (f_L). The lower of the two frequencies of a crystal unit in combination with a series load capacitor, under specified conditions, at which the electrical impedance of the crystal unit and capacitor is resistive. The value of f_L is related to f_r by the following expression:

$$f_L = f_r + (f_r/2)\ (C_1/(C_0+C_L))$$

where

C_1 is the motional capacitance
C_L is the load capacitor
C_0 is the crystal unit shunt capacitance

6.5.5 Resistance.

6.5.5.1 Resonance resistance (R_r). The resistance of the crystal unit alone at the resonant frequency, f_r.

6.5.5.2 Load resonance resistance (R_L). The resistance of the crystal unit in combination with a series load capacitor at the load resonance frequency, f_L. The value of R_L is related to the value of R_r by the following expression:

$$R_L = R_r(1+C_0/C_L)^2$$

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6.5.6 Frequency tolerances.

6.5.6.1 Overall tolerance. The maximum allowed deviation of the frequency (f_r or f_L as specified) from nominal frequency due to all specified causes. All environmental and electrical conditions must be specified as well as the time duration for which the tolerance is defined.

6.5.6.2 Aging tolerance. The permissible deviation due to time under specified conditions. The frequency change per unit of time (aging rate) under constant environmental conditions is, in general, not a constant. Therefore, extrapolation of aging must be performed with care, taking environmental changes into account.

6.5.7 Drive level. A measure of the electrical excitation of the crystal unit expressed in terms of the current through the device at a resonant frequency.

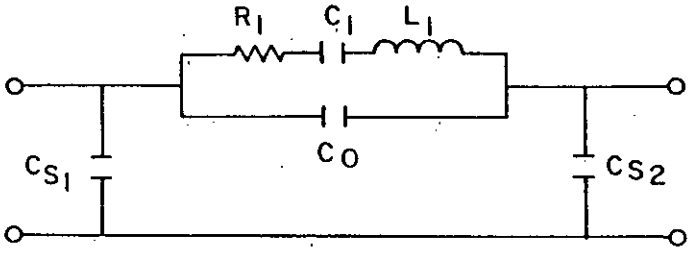
NOTE: Older specifications state the level of drive is the power dissipated by a calibration resistor.

6.5.8 Unwanted mode. A resonance of a crystal unit other than that associated with the nominal frequency. It should be noted that all crystal cuts have unavoidable modes of response. These modes must be considered when specifying the limit level of unwanted modes.

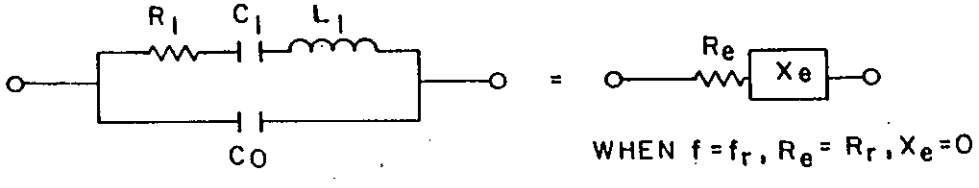
6.5.9 Coupled modes (frequency - resistance anomalies). Deviations from a normal frequency versus temperature characteristic. These deviations may be due to unwanted modes of vibration coupling to the desired mode, or they may be due to some processing or fabrication defect. The effect of coupled modes can generally be reduced by a reduction of crystal drive.

6.5.10 Equivalent circuits.

6.5.10.1 Three terminal equivalent circuit.

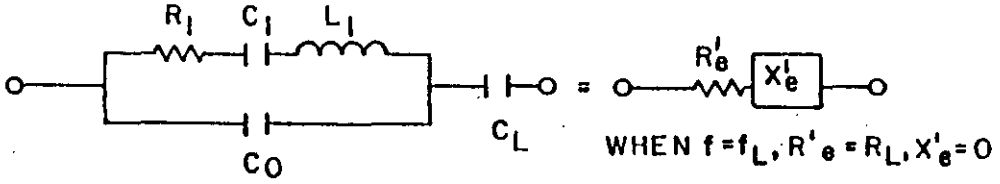


6.5.10.2 Two terminal equivalent circuit.



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6.5.10.3 Two terminal equivalent circuit with series load capacitor.



6.5.10.4 Shunt capacitance. C_0 is the capacitance between the two terminals of the resonator (see IEC-302).

6.5.10.5 Motional capacitance. The motional capacitance is C_1 (see IEC-302).

6.5.10.6 Stray capacitance. Cs1 and Cs2 are the stray capacitances of each terminal to the package.

6.5.11 Thickness shear resonator. A crystal unit using a thickness shear mode of motion of the quartz resonator.

6.6 Conditions for use of level B preservation. When level B preservation is specified (see 5.1.2), this degree of protection should be used for the acquisition of crystal units for resupply worldwide under known favorable handling, transportation and storage conditions.

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement, NATO NEPR 39. When amendment, revision, or cancellation of this specification is proposed, which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required. The United States, by international agreement NEPR 39, has agreed to the use of types of crystal units designated by NATO Nomenclature, i.e, NXT1/A, on new equipment design. These NATO types are shown in supplement 1 to this specification and should be used whenever possible.

6.9 Subject term (key word) listing.

Resonance frequency
Load resonance
Resonance resistance
Load resonance resistance
Aging
Unwanted modes
Coupled modes
Drive level
Motional capacitance
Thickness shear resonator
Shunt capacitance
Frequency range
Frequency tolerance
Drive sensitivity
Environmental tolerance

6.10 Manufacturers' codes. Manufacturers' codes are listed in publication cataloging handbook H4-1, "Commercial and Government Entity (CAGE)". All requests for this publication should be directed to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Ave., Philadelphia, PA 19120.

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APPENDIX A

PROCEDURE FOR QUALIFICATION INSPECTION FOR CLASS B CRYSTAL UNITS

10. SCOPE

10.1 Scope. This appendix details the procedure to submit samples with related data for qualification of class B crystal units covered by this specification. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS This section is not applicable to this appendix.

30. SUBMISSION.

30.1 Sample. Unless otherwise specified, 45 sample units of each crystal type, except for grouping permitted in table I, shall be submitted. If it is desired to qualify for a limited frequency band within the crystal unit range, 45 sample units shall be required in that band. Fifteen of the 45 unit group shall be at the lower end, 15 near the middle, and 15 at the upper end, of the frequency range over which qualification is sought.

30.2 Certification of material. When submitting samples for qualification, the supplier shall submit certification in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

40. EXTENT OF QUALIFICATION

40.1 Additional qualification. If a supplier qualifies for a limited frequency band within the frequency range of a crystal unit, corresponding limitation shall apply to the automatic qualification of other types. In no case shall the frequency range of the type receiving automatic qualification extend beyond the frequency range for which qualification has been obtained.

40.2 Frequency range. It is not necessary to qualify over the entire frequency range of a unit type; only that frequency range will be required for which the manufacturer has production capability. If, subsequently, a manufacturer wishes to qualify his product over a wider range of frequencies, he shall then request instructions from the Government qualifying activity.

40.3 Required tests. If the extension of the frequency range for the crystal type does not exceed 20 percent of the total range, only the following tests are required for qualification:

Drive sensitivity (of frequency and resistance)
Frequency and resistance.
Unwanted modes.
Seal

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APPENDIX B

GUIDELINES FOR THE IMPLEMENTATION OF CERTIFICATION OF CLASS S CRYSTAL UNITS

10. SCOPE.

10.1 Scope. This appendix provides guidelines to the certification team for the implementation of class S certification of manufacturing facilities. These guidelines will establish a uniform approach by the certification team in the assessment of the manufacturer's capability to consistently produce class S crystal units by providing instructions for assuring process control, applicable test methods, and records. A checklist is included for the use of the certification team (see table V). This appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. INSTRUCTIONS

30.1 Baseline documentation. At the time of the initial survey, a baseline shall be established. It is not intended that the manufacturer divulge proprietary information. Each manufacturer shall provide a detailed process flow chart which shall include as a minimum the following:

a. All major process steps and inspection points by descriptive title, document number, and revision.

b. The identification of all critical manufacturing and inspection steps as determined by the manufacturer in conjunction with the audit team at the time of baseline approval.

30.2 Changes to baseline documentation.

a. Minor changes. Any subsequent departure from the approved flow chart (i.e., additions or deletions), or changes to the revision status of those processes or inspections deemed critical are allowed provided they do not violate the changes prohibited by section b (major changes). These minor changes shall be described in general terms to the audit team during subsequent audits. This action shall result in an update of the baseline document.

b. Major changes. No changes (i.e., substitutions) shall be allowed in the following areas without prior review and approval of the qualifying activity. In these areas, partial or total requalification may be required as determined by the qualifying activity.

(1) Crystal unit characteristics.

(2) Bonding materials or techniques.

(3) Quartz material.

(4) Blank geometry.

(5) Electrode material or geometry.

(6) Case design.

(7) Sealing parameter - including backfill gas.

(8) Manufacturing methods and inspections.

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30.3 Audits. An initial audit and subsequent periodic audits (every 12 months or as necessary). The audits shall include, but not be limited to all of the items on the checklist plus MIL-STD-790 requirements. The result of any additional items that are assessed shall be added to the checklist for record purposes.

a. The audit team shall review the product assurance program plan and the manufacturing flow chart for adequacy and completeness prior to the audit of the manufacturing facilities and should be used as a baseline/reference during the course of the audit.

b. The audit team shall verify that all measuring and test equipment is functioning and calibrated. They shall also verify that test measuring procedures and equipment are sufficiently accurate for the required test.

c. For recertification, the audit team shall check 12 months of records for the item being recertified. In addition, the location of the equipment and the adequacy of the operating procedure shall be checked and whenever practical, operations utilizing the equipment and procedures witnessed.

TABLE V. Certification checklist procedure.

Check item 1/	Inspection	Satisfactory action
a. Flow chart.		
(1) Critical processes identified with revision.		
(2) Critical inspection identified with revision.		
(3) Nonproprietary documents furnished.		
(4) Screens/inspection points identified.		
b. Baseline shall describe as a minimum the following which shall include part numbers.		
(1) Part number/characteristics		
(2) Bonding material and techniques.		
(3) Quartz material.		
(4) Blank geometry.		
(5) Electrode material and geometry.		
(6) Sealing conditions.		
(7) Case.		
c. Lot control plan.		
d. Traceability		
e. Training plan for personnel.		
f. Nondestructive testing.		
g. Destructive testing.		
h. Verification of MIL-STD-790 requirements.		
i. Equipment calibration.		
j. Record maintainability.		
k. Reporting plan.		

1/ The audit team shall insure that adequate documentation, trained personnel, equipment, and visual aids are in evidence to perform the certification checklist procedure.

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APPENDIX C

PROCEDURE FOR QUALIFICATION INSPECTION FOR CLASS S CRYSTAL UNITS

10. SCOPE

10.1 Scope. This appendix details the procedure to submit samples with related data for qualification of class S crystal units covered by this specification. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. SUBMISSION

30.1 Sample. Unless otherwise specified, 45 sample units of each crystal type, except for grouping permitted in table I, shall be submitted. If it is desired to qualify for a limited frequency band within the crystal unit range, 45 sample units shall be required in that band. Fifteen of the 45 unit group shall be at the lower end, 15 near the middle, and 15 at the upper end, of the frequency range over which qualification is sought.

30.2 Certification of material. When submitting samples for qualification, the supplier shall submit certification in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

40. EXTENT OF QUALIFICATION

40.1 Additional qualification. If a supplier qualifies for a limited frequency band within the frequency range of a crystal unit, corresponding limitation shall apply to the automatic qualification of other types. In no case shall the frequency range of the type receiving automatic qualification extend beyond the frequency range for which qualification has been obtained.

40.2 Frequency range. It is not necessary to qualify over the entire frequency range of a unit type; only that frequency range will be required for which the manufacturer has production capability. If, subsequently, a manufacturer wishes to qualify his product over a wider range of frequencies, he shall then request instructions from the Government qualifying activity.

40.3 Required tests. If the extension of the frequency range for the crystal type does not exceed 20 percent of the total range, only the following tests are required for qualification:

Drive sensitivity (of frequency and resistance).
Frequency and resistance.
Unwanted modes.
Seal

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APPENDIX D

PREPARATION GUIDELINES FOR SPECIFICATIONS

10. SCOPE

10.1 Specification guidance. This section is included for guidance in generating new specification sheets. Recommendations are made for various testing parameters. Sample format is included. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. Requirement categories.

30.1 Basic requirements. General requirements common to all crystal units of military quality. All crystal units in compliance with this specification shall be subjected to the inspections listed in table VI.

30.2 Special requirements. Requirements of a more specialized nature than the basic requirements (see 30.1) which may be required for some applications. Special requirements for which a test procedure has been defined are listed in table VII. Other requirements may be imposed, but a test procedure must then be included.

30.3 Seal leak rates. The following leak rates may be specified:

level	Leak rate (atm cm^3s of air)
A	1×10^{-8}
B	1×10^{-9}
C	1×10^{-10}
D	as specified, but no less than 1×10^{-8}

30.4 Frequency versus temperature parameters (see 3.10). The parameters to be within the specified tolerances for the f versus T test shall be selected from the following list:

a. Frequency at any temperature in the specified range
b. UTP (see 30.4.1.2)
c. LTP (see 30.4.1.3)
d. Ti (see 30.4.1.7)
e. f(UTP) (see 30.4.1.4)
f. f(LTP) (see 30.4.1.5)
g. f(Ti) (see 30.4.1.8)
h. Slope at temperature T, df(T)/dT
i. Slope at the inflection temperature, df(Ti)/dT
j. Slope at any temperature in the specified range
k. f(UTP)-f(LTP)
l. Frequency within a tolerance in a specified temperature range

30.4.1 Frequency versus temperature definitions. See figure 6.

30.4.1.1 Turnover point. The point, frequency and temperature, at which the slope (i.e., first derivative) of the frequency versus temperature characteristic is equal to zero.

30.4.1.2 Upper turnover temperature, UTP. The higher of two turnover temperatures.

30.4.1.3 Lower turnover temperature, LTP. The lower of two turnover temperatures.

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30.4.1.4 F(UTP). The frequency at the upper turnover temperature.

30.4.1.5 F(LTP). The frequency at the lower turnover temperature.

30.4.1.6 Inflection point. The point, frequency and temperature, at which the curvature (i.e., second derivative) of the frequency versus temperature characteristic is equal to zero.

30.4.1.7 Inflection temperature, Ti. The temperature of the inflection point.

30.4.1.8 f(Ti). The frequency at the inflection temperature.

30.4.2 Rate of change of temperature. The rate of change of temperature for the f versus T test shall be selected from the following list:

a. 0.5°C/min or less
b. 1.0°C/min "
c. 5.0°C/min "
d. 10°C/min "
e. As specified (see 3.1)

30.4.3 Data recording interval. The data recording interval for the f versus T test shall be selected from the following list:

a. Continuous
b. 0.2°C
c. 0.5°C
d. 1.0°C
e. 3.0°C
f. As specified (see 3.1)

30.5 Frequency and resistance offset, vibration (see 3.19.1). The purpose of this test is to determine the permanent effect of a vibratory environment which may be encountered during shipment or use.

30.5.1 Test conditions for method 204. If method 204 of MIL-STD-202 is specified, the test condition shall be selected from the following list:

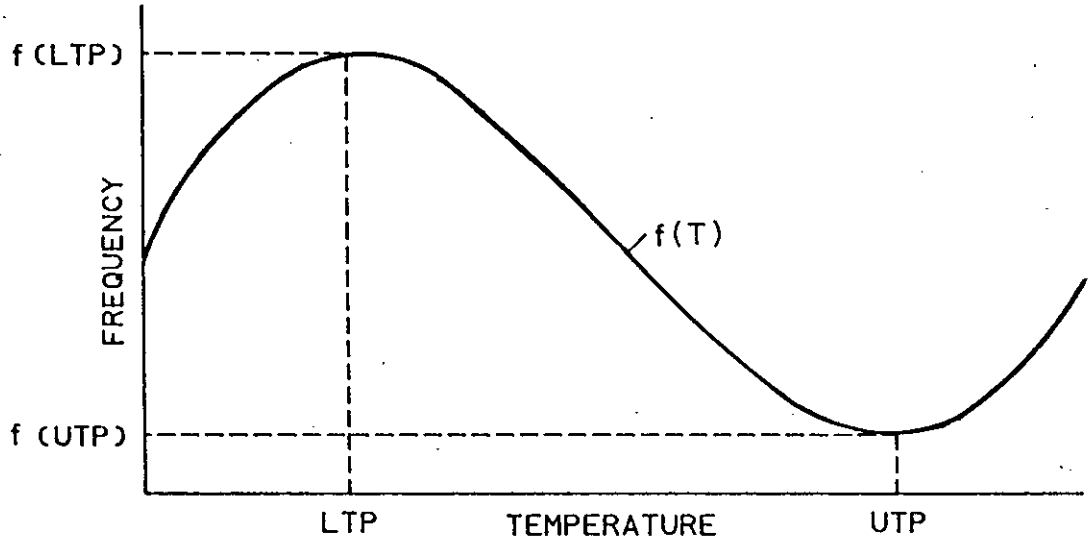
Method 204 - swept sine vibration

1. A (10 g peak, 10-500 Hz)
2. B (15 g peak, 10-2000 Hz)
3. C (10 g peak, 55-2000 Hz)
4. D (20 g peak, 10-2000 Hz)
5. E (50 g peak, 10-2000 Hz)
6. F (20 g peak, 10-3000 Hz)
7. G (30 g peak, 10-2000 Hz)
8. As specified (see 3.1) but no less than condition A.

30.5.2 Test condition for method 214. If method 214 of MIL-STD-202 is specified, the test condition shall be selected from the following list:

Method 214 - random vibration
1. I (letter G, 23.1g rms)
2. As specified (see 3.1)

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NOTE: Slope: $S(T) = df(T)/dT$
Curvature: $K(T) = dS(T)/dT$
Turnover point: $S(T) = 0$
Inflection point: $K(T) = 0$

FIGURE 5. Frequency versus temperature definitions.

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30.6 Acceleration sensitivity (vibration) (see 3.19.2). The purpose of this test is to determine the instantaneous effect of vibration on an operating crystal unit. The method of measurement consists of the determination of the relative magnitudes of the vibration induced sidebands and the carrier.

30.7 Frequency and resistance offset (steady state acceleration) (see 3.19.3). The purpose of this test is to determine the permanent effects of a constant acceleration such as would be experienced in missile launch.

30.8 Acceleration sensitivity (steady state) (see 3.19.4). The purpose of this test is to determine the instantaneous effect of a constant acceleration such as would be experienced during missile launch.

30.9 Acoustical noise (see 3.19.5). The purpose of this test is to determine the instantaneous effect of a high sound pressure level on the frequency of the crystal unit.

30.10 Thermal shock (see 3.26). The purpose of this test is to determine the permanent effect of a sudden temperature change.

30.11 Thermal time constant (see 3.27). The purpose of this test is to determine the time it takes for a crystal unit to reach equilibrium after a sudden change in temperature.

30.11.1 Dynamic f versus T characteristic (see 3.27). The frequency versus temperature characteristic measured during the thermal time constant test, i.e., after a step temperature change, is called the dynamic f versus T characteristic.

30.12 Frequency overshoot (see 3.27.1). The purpose of this test is to measure the thermal stress sensitivity of the crystal unit.

30.13 Thermal frequency repeatability (see 3.20). The purpose of this test is to measure the frequency shifts due to thermal cycling.

30.14 Radiation hardness (see 3.32). The purpose of this test is to determine the permanent and temporary effects of ionizing radiation.

30.15 Coupled modes (see 3.10.3). The purpose of this test is to determine if any vibrational modes of the crystal unit other than the desired mode are being excited at specific temperatures. These coupled modes will drain acoustic energy from the desired mode and possibly cause a resistance increase, cessation of oscillation, and distortion of the f versus T curve.

30.16 Bond strength (see 3.25). The purpose of this test is to determine, prior to sealing, if the strength of the bond between the resonator blank and the mounting structure is sufficient for the intended application.

30.17 Accelerated aging (see 3.15). The purpose of this test is to act as a pre-delivery quality control procedure. This test cannot be used as an aging predictor.

30.18 Aging (see 3.31). The purpose of this test is to determine the effects of time on the frequency and resistance of the crystal unit.

30.19 Drive sensitivity (of frequency and resistance) (see 3.16). The purpose of this test is to act as an indicator of poor processing of the resonator. Loose particles or other surface defects will cause high resistance at low levels of drive. This test was instituted as a substitute for the blank surface finish requirements in previous crystal specification.

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30.20 Short-term-stability (see 3.30). The purpose of this test is to measure the "noise" of the crystal unit. The measure being used is the Allan Variance, which is a time domain type of measurement.

30.21 Condensibles (see 3.17). The purpose of this test is to identify volatile materials, such as water vapor or flux residue, inside the crystal enclosure.

30.22 Quality factor (see 3.12). The purpose of this test is to provide an alternative to the specification of resistance and motional capacitance. Since Q is a function of the motional capacitance and the resistance at a given frequency, only two of the three parameters, "Q", motional capacitance and resistance, can be independently specified.

30.23 Particle impact noise detection (PIND) (see 3.21). PIND testing is effective in defecting loose particles, however pre- and co- shock levels can damage crystals and should be set at a nondegrading level for the specific device being tested.

TABLE VI. Basic requirements.

Inspection	Requirement paragraph	Test method paragraph
Materials	3.5	4.3
Holder - base and cover	3.5.1	
Design and construction	3.6	4.9.2
Supporting structure	3.6.1	
Bonding	3.6.2	
Sealing conditions	3.6.3	
Glass parts	3.6.4	
Pin alignment	3.6.5	
Final frequency adjustment	3.6.6	
Solderability or lead attachment	3.7	4.9.3
Solderability 1/	3.7.1	4.9.3.1
Ceramic package 1/	3.7.2	4.9.3.2
Resistance to solvents	3.8	4.9.4
Frequency and resistance	3.10	4.9.6
Resistance versus temperature (R versus T)	3.10.1	4.9.6.1
Frequency versus temperature (static)	3.10.2	4.9.6.1
Coupled modes (frequency-resistance anomalies)	3.10.3	4.9.6.2
Unwanted modes	3.13	4.9.9
Insulation resistance	3.14	4.9.10
Accelerated aging	3.15	4.9.11
Drive sensitivity (of frequency and resistance)	3.16	4.9.12
Condensibles	3.17	4.9.13
Shock	3.18	4.9.14
f and R offset, vibration	3.19.1	4.9.15.1
Moisture resistance	3.22	4.9.18
Salt spray	3.23	4.9.19
Terminal strength	3.24	4.9.20
Terminal pull	3.24.1	4.9.20.1
Wire-lead bend	3.24.3	4.9.20.2
Wire-lead twist	3.24.4	4.9.20.4
Thermal shock, visual damage	3.26	4.9.22
Seal	3.29	4.9.25
Aging	3.31	4.9.27
Marking	3.33	4.9.2.1
Workmanship	3.34	4.9.2

1/ When applicable

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TABLE VII. Special requirements.

Inspection	Requirement paragraph	Test method paragraph
Resistance to soldering heat	3.9	4.9.5
Capacitance	3.11	4.9.7
Shunt	3.11.1	4.9.7.1
Motional	3.11.2	4.9.7.2
Quality factor ("Q")	3.12	4.9.8
Vibration sensitivity	3.19.2	4.9.15.2
Frequency and resistance offset (steady state acceleration)	3.19.3	4.9.15.3
Acceleration sensitivity (steady state)	3.19.4	4.9.15.4
Acoustical noise	3.19.5	4.9.15.5
Thermal frequency repeatability	3.20	4.9.16
Thermal frequency hysteresis	3.20.1	4.9.16.2
PIND	3.21	4.9.17
Bond strength	3.25	4.9.21
Thermal shock, f and R offset	3.26	4.9.22
Thermal time constant	3.27	4.9.23
Frequency overshoot	3.27.1	4.9.23.1
Barometric Pressure	3.28	4.9.24
Short term stability	3.30	4.9.26
Radiation hardness	3.32	4.9.28
Total dose	3.32.1	4.9.28.1
Dose rate	3.32.2	4.9.28.2
Neutrons	3.32.3	4.9.28.3
Accumulated time error	3.32.4	4.9.28.4

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Tabular index for creation of new specification sheets; creation of a new specification sheet requires first article inspection. The creation of a new crystal type requires qualification inspection.

Frequency range: Overtone: Intended purpose:

Inspection	Requirement paragraph	Test method paragraph	Inspection variables	Acceptance criteria
Materials	3.5	4.3	Certification required (Y/N)	Required certificate
Holder	3.5.1	4.9.2	Type_____	
Quartz	3.5.2	4.3	Cultured (Y/N) Q-grade_____ Grower (Y/N) Autoclave ID (Y/N) Date code (Y/N) Bar ID (Y/N) Swept (Y/N) Sweeper ID (Y/N) Sweep lot (Y/N) Electrode desc. (Y/N) Atmosphere (Y/N) Temperature (Y/N) Electric field (Y/N) Current profile (Y/N)	Required certificate
Sealing conditions	3.6.3	4.3	Vacuum (Y/N) Backfill gas_____	Required certificate
Ceramic package lead attachment	3.7.2	4.9.3.2	Bond method T/C (Y/N) Weld (Y/N)	4.9.3.2
Seal	3.29	4.9.24		Leak rate<_____
Resistance versus temperature	3.10.1	4.9.6.1	Low temperature limit_____C High temperature limit_____C C(load)_____pF Drive current_____ma	R<_____
Frequency versus temperature	3.10.2	4.9.6.1	Low temperature limit_____C High temperature limit _____pF C(load)_____pF Drive current_____ma dT/dT maximum_____ Data interval_____ dT/dt (neat TP)_____ Data Int. (TP)_____	f_____ ± _____ f(UTP)___ ± _____ f(LTP)___ ± _____ f(Ti)___ ± _____ UTP_____ ± _____ LTP_____ ± _____ Ti_____ ± _____ df/dT max._____ df/dT min._____

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Inspection	Requirement paragraph	Test method paragraph	Inspection variables	Acceptance criteria
Coupled modes	3.10.3	4.9.6.2	Low temperature limit_____C High temperature limit_____C	f/f<_____ R<_____
Shunt capacitance	3.11	4.9.7.1		C_o_____ ± _____
Motional capacitance	3.11.2	4.9.7.2		C_i_____ ± _____
Quality factor "Q"	3.12	4.9.8		Q>_____
Aging	3.31	4.9.27	Aging temperature___C	f/f(30 days)_____ df/dt(day 30)_____ f(max 24 hrs.)_____ max variance_____ R<_____
Accelerated aging	3.15	4.9.11		f/f<_____ R_____
Drive sensitivity	3.16	4.9.12	Minimum current____ua Maximum current____ma	R_____ f/f<_____
Unwanted modes	3.13	4.9.9	Minimum frequency Maximum frequency	R_____
Short term stability	3.30	4.9.26		See figure
Shock	3.18	4.9.14	Acceleration level___ Pulse shape_____	f/f<_____ R<_____
Vibration frequency and resistance offset	3.19.1	4.9.15.1	MIL-STD-202: Method 204 (Y/N) Peak acceleration ____ Minimum frequency____ Maximum frequency____ Duration_____ or Method 214 (Y/N) rms g level_____ Duration_____	f/f<_____ R<_____
Vibration sensitivity	3.19.2	4.9.15.2	Maximum acceleration_____ Minimum vibration frequency_____ Maximum vibration frequency_____	Gamma<_____ Gamma<_____

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APPENDIX D

Inspection	Requirement paragraph	Test method paragraph	Inspection variables	Acceptance criteria
Acceleration frequency and resistance offset	3.19.3	4.9.15.3	Maximum acceleration duration_____	f/f_____ R<_____
Acceleration sensitivity	3.19.4	4.9.15.4	Maximum acceleration_____	f/f_____
Acoustical noise	3.19.5	4.9.15.5	MIL-STD-810 Method 515: Category_____ Overall sound pressure_____ Exposure time_____	
Thermal shock frequency and resistance offset	3.26	4.9.22		f/f<_____ R<_____
Thermal time constant	3.27	4.9.23		tau<_____
Frequency overshoot	3.27.1	4.9.23.1		f/f<_____
Thermal frequency repeat-ability	3.20	4.9.16	Number of cycles_____	f/f(UTP)<_____ f/f(LTP)<_____ (cf)<_____
Thermal frequency repeat-ability	3.20	4.9.16	Number of cycles_____	f/f(UTP)<_____ f/f(LTP)<_____ (cf)<_____
Thermal frequency hysteresis	3.20.1	4.9.16.1	Low temperature limit_____ High temperature limit_____ Number of cycles_____	f/f<_____
Radiation hardness total dose	3.32.1	4.9.28.1	Dose_____	f/f<_____ R<_____

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APPENDIX D

Inspection	Requirement paragraph	Test method paragraph	Inspection variables	Acceptance criteria
Radiation hardness dose rate	3.32.2	4.9.28.2	Pulse level Duration Anneal time Maximum frequency Duration	f/f R post anneal f/f R
Radiation hardness neutrons	3.32.3	4.9.28.3	Fluence	f/f R
Radiation hardness accumulated time error	3.32.4	4.9.28.4	Total dose Dose rate	t
Bond strength	3.25	4.9.21		Bond strength
Moisture resistance f and R offset	3.22	4.9.18		f/f R
PIND sensitivity	3.21	4.9.17	Pre-shock level Co-shock level	

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APPENDIX E

ZERO DEFECT SAMPLING TABLES

10. SCOPE

10.1 Scope. This appendix details the tables to be used for group A and B inspections of crystal units covered by this specification. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. REQUIREMENTS.

30.1 Basic requirements. Tables VIII and IX shall be used as they are applicable and referenced in section four of MIL-C-49468.

TABLE VIII. Group A, zero defect sampling plan.

Lot size	Sample size		
	Class S	Class B General 1/	
		Internal	External
1-8	All	All	3
9-15	All	13	3
16-25	All	13	3
26-50	All	13	5
51-90	All	13	6

1/ If lot size is smaller than sample size, test all of the units. One or more sample rejects shall be cause for failure of the lot.

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APPENDIX E

TABLE IX. Group B, zero defect sampling plan.

Lot size	Class S		Class B	
	Subgroup 1	Subgroup 2	Subgroup 1	Subgroup 2
1 - 500	21	29	16	21
501 - 12000	27	34	19	27
12001 - 32000	35	42	23	35
32000 and over	38	50	29	38

One or more sample rejects shall be cause for failure of the lot.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - CG, MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0649)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-49468	900914

3. DOCUMENT TITLE
Crystal Units, Quartz, Precision, General Specificaiton For

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander LABCOM	b. TELEPHONE (Include Area Code) (1) Commercial (908)544-3441	(2) AUTOVON 995-3441
c. ADDRESS (Include Zip Code) ATTN: SLCET-RS Ft. Monmouth, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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